



SAUDI ARABIAN NATIONAL GEOSPATIAL ECOSYSTEM

General Authority for Survey and Geospatial Information

Note

This document has been developed in collaboration between the General Authority for Survey and Geospatial Information (GEOSA) & The Open Geospatial Consortium (OGC). Both organizations have used their best of the ability to crosscheck the content. Any omissions or errors may please be notified.

Contents:

List of Figures	1
List of Abbreviations	1
1 Executive Summary	3
2 About GEOSA	5
3 Geospatial Ecosystem Overview	7
4 Geospatial Ecosystem’s Aligning with UN-IGIF and SDGs	16
5 Saudi Arabian National Geospatial Ecosystem Requirements	26
6 Saudi Arabian National Geospatial Ecosystem Framework	29
7 Saudi Arabian National Geospatial Ecosystem Framework	35
7 Saudi Arabian National Geospatial Ecosystem Initiatives	39
8 References	66



List of Figures

Figure 1	GEOSA Objectives	6
Figure 2	UN-GGIM Resolutions on Geospatial Ecosystem	8
Figure 3	Geospatial Ecosystem	11
Figure 4	Geospatial Ecosystem Elements	12
Figure 5	Emerging Conditions Impacting Geospatial Ecosystem	13
Figure 6	Features of Geospatial Ecosystem	15
Figure 7	Nine Strategic Pathways of the UN-IGIF	17
Figure 8	Geospatial Ecosystem Characteristics	22
Figure 9	Sustainable Development Goals	26
Figure 10	Integrated Geospatial Information required for achieving Sustainable Development Goals	27
Figure 11	Global Fundamental Geospatial Data Themes most relevant to the SDGs	28
Figure 12	Factors influencing Saudi Arabian Geospatial Ecosystem	29
Figure 13	Saudi Arabian National Geospatial Ecosystem high-level components	30
Figure 14	National Geospatial Ecosystem Vision, Mission, Values and Functions for Geospatial Ecosystem	37
Figure 15	Functions of the Geospatial Ecosystem	38
Figure 16	National Geospatial Ecosystem Vision, Mission, Values, Functions, Initiatives and Pathways for Geospatial Ecosystem	39
Figure 17	Breakdown of Saudi Arabian National Geospatial Ecosystem Initiatives	40
Figure 18	Saudi Arabian National Geospatial Ecosystem Initiatives	41
Figure 19	Initiative 1 Overview	42
Figure 20	Initiative 2 Overview	47
Figure 21	Initiative 3 Overview	51
Figure 22	Initiative 4 Overview	55
Figure 23	Initiative 5 Overview	59
Figure 24	Initiative 6 Overview	62

List of Abbreviations

AI/ML	Artificial Intelligence/Machine Learning
API	Application Programming Interface
AR	Augmented Reality
DCAT	Data Catalog
DCAT-AP	Data Catalog Application Profile
DCAT-PROV	Data Catalog Provenance
DSBA	Data Spaces Business Alliance



DSSC	Data Spaces Support Centre
ECOSOC	Economic and Social Council
EOSC	European Open Science Cloud
FAIR	Findable, accessible, interoperable and reusable
G2B	Government to Business
G2C	Government to Citizens
G2G	Government to Government
GBIF	Global Biodiversity Information Facility
GeoDCAT	Geo- Data Catalog
GEOSA	General Authority for Survey and Geospatial Information
GEOSS	Global Earth Observation System of Systems
GKI	Geospatial Knowledge Infrastructure
GNSS	Global Navigation Satellite Systems
HMI	Human Machine Interface
HPC	High-Performance Computing
IGIF	Integrated Geospatial Information Framework
IoT	Internet of Things
ISO/TC	International Organization for Standardization/Technical Committee
JSON-LD	JavaScript Object Notation for Linked Data
ISDA	International Data Spaces Association
LD	Linked Data
M2M	Machine-to-Machine Communication
NGC	National Geospatial Centre
OGC	Open Geospatial Consortium
OLAP	Online Analytical Processing
QA	Quality Assurance
QC	Quality Control
RPA	Robotic Process Automation
SDG	Sustainable Development Goals
SDI	Spatial Data Infrastructure
SoS	System of Systems
SKI	Spatial Knowledge Infrastructure
UN-GGIM	United Nations Committee of Experts on Global Geospatial Information Management
URI	Unique Resource Identifier
VR	Virtual Reality
XML	Extensible Markup Language



Executive Summary

The Kingdom of Saudi Arabia stands at the forefront of embracing change during unprecedented global transformations in environmental, social, economic, and technological domains. The dynamic landscape has spurred a significant evolution in the field of Geospatial information, and Saudi Arabia, under the ambitious banner of Saudi Vision 2030, is strategically aligning itself with this paradigm shift. Recognizing the pivotal role of digital and geospatial technologies in fostering a sustainable and efficient economy, the Kingdom's General Authority for Survey and Geospatial Information (GEOSA) has laid the foundation for the world's first Geospatial Ecosystem, called the **Saudi Arabian National Geospatial Ecosystem**.

Supporting the Integrated Geospatial Information Framework (IGIF) of the United Nations Committee of Experts on Global Geospatial Information Management (UN-GGIM), this comprehensive initiative reflects a commitment to addressing the challenges posed by socioeconomic dynamics, digital advancements, and environmental shifts.

The Integrated Geospatial Information Framework (IGIF) is a broad-level framework that establishes core principles for geospatial information management. The Geospatial Ecosystem provides the necessary rules, regulations, and procedures to implement the IGIF, achieving socio-economic benefits. The Geospatial Ecosystem focuses on creating a solid, integrated, and secure geospatial and IT infrastructure that supports automation, innovation, intelligent analytics, and collaboration between diverse stakeholders to share and use data.

The Saudi Arabian National Geospatial Ecosystem will be operated by the National Geospatial Centre (NGC) at the General Authority for Survey and Geospatial Information (GEOSA) aimed at enhancing geospatial capacities within the nation. By setting standards for the proper use of geospatial data and fostering collaboration, the NGC endeavors to make geospatial information accessible, secure, and adaptable to ongoing developments. This forward-looking approach positions the Saudi Arabian National Geospatial Ecosystem as a sustainable, active, and nationally relevant framework, intertwining people, policies, practices, and procedures to meet the demands of the modern world. This transformative journey is a testament to Saudi Arabia's commitment to harnessing geospatial knowledge in shaping a resilient and responsive future.

The Saudi Arabian National Geospatial Ecosystem (the Geospatial Ecosystem) is a comprehensive and interconnected system that uses full-spectrum interoperability technology and Artificial Intelligence to adapt dynamically to changing conditions. It is user-centric and provides orthogonal services, such as cyber security, governance, legal security, and trust-building services. The Ecosystem will continuously adapt to the environmental changes to advance as a technology framework.

To effectively achieve the objectives of the Saudi Arabian National Geospatial Ecosystem, the National Geospatial Centre (NGC) will adopt new functions that will enable it to embrace the Geospatial Ecosystem concept and its associated benefits fully. These new functions will focus on **Geospatial Data, Geospatial Intelligence, National Geospatial Platforms, Geospatial Governance, and Geospatial Security**.

The Saudi Arabian National Geospatial Ecosystem is structured around six high-level **Initiatives** with 30 associated **Pathways**, that culminate in achieving 51 **Outputs**.

The initiatives provide a direction for the development and implementation of the Ecosystem, and at their core, address the most immediate and practical concerns related to making the



data and systems currently adopted in NGC Systemic, Dynamic, User-Centric, Knowledge-Based, Automated, Secure, Collaborative, Competitive, and Futuristic.

The successful implementation of the Initiatives relies on the execution of one or more Pathways, specifically designed to attain the outcomes prescribed for each Initiative. Pathways essentially represent a task or process to be undertaken when implementing the geospatial ecosystem around the Initiatives. Pathways would result in producing capabilities or functionalities otherwise referred to as Outputs that would characterize and operationalize the ecosystem.

The successful establishment and delivery of the Initiatives are crucial for NGC to transform its data infrastructure to a national ecosystem and achieve its Mission Statement: Saudi Arabian National Geospatial Ecosystem provides a ubiquitous network of interconnected Geospatial information, products, technology, and services, enabling the delivery of knowledge-based insights to facilitate a bottom-up approach to problem-solving and decision-making.

INITIATIVE 1: ENABLE GEOSPATIAL ECOSYSTEM PRACTICES

The Initiative aims to transform the Geospatial ecosystem into a thriving, interconnected hub where stakeholders collaborate seamlessly, data flows freely, and Geospatial services play a central role in addressing modern-day challenges. Through the Pathways, it seeks to establish a foundation for a more robust and responsive geospatial landscape that serves the broader community. Under this Initiative, the NGC will be developing frameworks for various kinds of partnerships with stakeholders, nationally, regionally and internationally. Apart from these, the Initiative will establish policies and frameworks for data access and use, provide training and develop a geospatial ecosystem maturity framework.

INITIATIVE 2: OPERATIONALIZE GEOSPATIAL ECOSYSTEM SHARED INFRASTRUCTURE

The Initiative seeks to build and operate the underlying technical infrastructure required to enable the various geospatial functionalities promoted by the Geospatial Ecosystem, ranging from spatial data access analysis to visualization and dissemination. The focus of this Initiative will be to build a geospatial cloud management system, high-performance and distributed computing as a service, API and reliable hardware, software, and infrastructure.

INITIATIVE 3: ENABLE GEOSPATIAL ECOSYSTEM DATA MANAGEMENT

The Initiative is a strategic effort to establish comprehensive, efficient, and structured data management practices within the Geospatial Ecosystem to promote the development of comprehensive, well-structured, and efficient data management practices by incorporating relevant policies, standards, and guidelines. The focus of this Initiative will be on developing and implementing a geospatial data interoperability model, licensing model, semantics and ontologies, and quality assurance framework.

INITIATIVE 4: ENHANCE GEOSPATIAL ECOSYSTEM DATA ANALYTICS

This Initiative will enable the delivery of spatially driven insights to assist in real-world



decision-making by users and stakeholders. This would involve Big Data Analytics, Spatial Analysis workflows dealing with raster and vector datasets, and visualization and simulation as an added medium of insights generation. The Initiative would provide ready-to-apply data analytics and allow users to perform data analysis themselves. The focus of the Initiative will be on creating an ecosystem analytical resource library, developing, and implementing analytics interoperability framework, advanced visualization platforms for various applications, and real-time and distributed analytics solutions delivery.

INITIATIVE 5: AUTOMATE GEOSPATIAL ECOSYSTEM FUNCTIONALITIES

This initiative aims to integrate a digital workforce through automation techniques and technologies. It will optimize standardized and repeatable processes, allowing data to be analyzed on a larger scale while minimizing human intervention. The initiative will establish a dynamic infrastructure that evolves constantly, ensuring that the Saudi Arabian National Geospatial Ecosystem remains at the forefront of innovation in using Artificial Intelligence and associated paradigms. This will enhance and enable geospatial workflows independent of human involvement. The focus of the Initiative will be to implement robotic process automation capabilities, develop a machine-to-machine communication(M2M) interface, and an Artificial Intelligence/Machine Learning (AI/ML) automation library.

INITIATIVE 6: DELIVER AN INTEGRATED GEOSPATIAL ECOSYSTEM ARCHITECTURE

The Initiative lays the foundation of architectural frameworks and working modalities for enabling all the aspired functionalities and capabilities of Saudi Arabian National Geospatial Ecosystem. It aims to establish a National Geospatial Ecosystem architecture that promotes integration and interoperability by connecting its business, data, application, technology, service delivery, and security components within the ecosystem. It is critical to facilitate integration, interoperability, secure data exchange, and resilience within the geospatial ecosystem. The focus will be on developing a national geospatial platform, implement user feedback mechanisms and loops, establish geospatial security and regulatory framework, threat detection module, increase Ecosystem Geospatial security awareness, and a backup and recovery framework.

Through the operationalization of the Saudi Arabian Geospatial Ecosystem, the Kingdom is taking a comprehensive approach to managing its geospatial assets and data and utilizing them more effectively. This serves as an example for the rest of the world to follow, particularly in supporting the United Nations' Integrated Geospatial Information Framework. The Geospatial Ecosystem will help the Kingdom achieve its national commitments, including the Sustainable Development Goals, and support the vision for Saudi Arabia in 2030.

About GEOSA

The Kingdom of Saudi Arabia has launched Vision 2030, an ambitious and transformative plan to reduce its reliance on oil revenues and diversify its economy. One of the critical initiatives of the Saudi Vision 2030 is adopting digital and geospatial technologies to create a more sustainable and efficient economy. Saudi Arabia has taken significant strides toward achieving



its digital goals. Together with the globally leading organization on geospatial standards, the Open Geospatial Consortium, the Kingdom has started to build a new generation of geospatial data exchange environment that goes far beyond the capacities of existing national and international spatial data infrastructures - **The Saudi Arabian National Geospatial Ecosystem.**

To build an efficient and sustainable digital future, it is important to understand geospatial data in broader contexts; thus, using the term “ecosystem” instead of infrastructure is strategic. The Saudi Arabian National Geospatial Ecosystem acknowledges that geospatial data does not live in isolation. Combining it with data from other sectors, such as social, economic, or governance, is necessary to exploit its full potential. And finally, acknowledging that data alone is not of value, unless it renders intelligent analytics that saves time, resources, and lives. Thus, the Geospatial Ecosystem ensures that integrated data from various sources yields automated analytics that helps those in decision-making position to take timely action.

As the national agency, the General Authority for Survey and Geospatial Information (GEOSA) regulates, develops, supervises, and monitors the Sector in the Kingdom in a manner that achieves quality, enhances performance, and maintains security in coordination with relevant agencies. The Authority has been tasked with undertaking any measure it deems necessary to achieve its objectives, particularly around achieving quality, maintaining performance, and maintaining security.



Figure 1: GEOSA Objectives

Achieving Quality: In order to promote growth and attract investments in the geospatial sector, GEOSA focuses on effective collaboration and coordination with relevant national and international organizations and counterparts. As a representative of the Kingdom, GEOSA takes up issues related to geospatial information on both local and international platforms. The organization works towards building and strengthening national capacities in the sector through partnerships with universities and specialized institutes within and outside the Kingdom. GEOSA develops and executes strategic plans while conducting relevant research and studies in collaboration with other relevant agencies.

Enhancing Performance: GEOSA is responsible for developing and approving the national geospatial infrastructure, geodetic reference, geodetic networks, and marine hydrographic



surveys in the Kingdom. They provide and market geospatial data, products, services, applications, and Atlases, as well as consult on land and maritime borders. In coordination with relevant agencies, GEOSA delivers databases and services for these purposes.

Maintaining Security: GEOSA is responsible for establishing and updating laws, rules, and standards to achieve integrated use of the geospatial information system while ensuring compliance. Their main goal is to maintain the security and confidentiality of geospatial data products and services. They also provide licenses for geospatial activities, oversee the training and classification of practitioners, and establish relevant rules to ensure proper functioning of the system.

To improve access to its data, GEOSA has taken significant steps recently. One such initiative was establishing the National Geospatial Centre (NGC), aimed at building Geospatial capacities within the Kingdom of Saudi Arabia. The National Geospatial Centre is responsible for governing the use of national geospatial data. They establish standards and criteria for the proper use of this information, and organize the collection, sharing, exchanging, saving, updating, maintaining, and publishing of geospatial data on the National Geoportal Platform. The principle followed is to make this data easily accessible, secure, and subject to updates and growth. The Centre also supports and enhances geospatial data analytics and helps identify strategic investments to expand these capabilities.

The other key initiative that GEOSA has committed to is developing the world's first Geospatial Ecosystem called the **Saudi Arabian National Geospatial Data Ecosystem**. Underpinned by a solid framework provided by the United Nations, the Geospatial Ecosystem is expanding GEOSA's perspective on how geospatial information is managed, shared, and utilized responding to the increasing need for timely, authoritative, and accurate data that supports smart insights and analytics. GEOSA is embarking on this exciting new journey by utilizing available technologies and responding to economic and social demands through the National Geospatial Center.

The Saudi Arabian National Geospatial Ecosystem is a sustainable, active, and nationally relevant Geospatial Ecosystem that will contribute to all activities and applications requiring any element of Geospatial knowledge. It will be served through an interconnected Geospatial Ecosystem of people, policies, practices, and procedures that address the needs of the modern world.

Geospatial Ecosystem Overview

The field of Geospatial information has experienced significant transformations in the last decade, marked by advancements in technology, an increase in the production of Geospatial information, the growing importance of location-based services, rapidly changing socio-economic and environmental conditions that demand high-quality Geospatial information, and the emergence of many entities that provide Geospatial-based solutions.

The rapid advancements have warranted professionals to acquire new skills to take advantage of these opportunities while appropriately managing data, protecting privacy, and addressing issues related to cybersecurity and bias. At the same time, the use of artificial intelligence/



machine learning (AI/ML) for automation and analytics has seen a significant rise, influencing various organizations' roles, relevance, and capabilities.

Saudi Arabia has responded to the changing times by adopting Saudi Vision 2030, which is an ambitious plan aimed at reducing the country's reliance on oil revenues and diversifying its economy. The vision focuses on the adoption of digital and geospatial technologies to create a sustainable and efficient economy. To support its focus, the Kingdom has heavily invested in creating a robust digital infrastructure, which includes high-speed internet, 5G networks, and cloud computing capabilities to facilitate digital services and applications like e-commerce, fintech, telemedicine, and distance learning.

In response to the socio-economic, digital, and environmental changes, the Kingdom's General Authority for Survey and Geospatial Information (GEOSA) has established the Saudi Arabian National Geospatial Ecosystem. This ecosystem will be active, sustainable, and nationally relevant, contributing to all activities and applications requiring geospatial knowledge.

UN-GGIM Resolutions on Geospatial Ecosystem

The United Nations Committee of Experts on Global Geospatial Information Management (UN-GGIM) plays a pivotal role in shaping the global discourse on geospatial information. At the 11th, 12th, and 13th sessions of the UN-GGIM active discussions surrounding the development and enhancement of the geospatial ecosystem took place.

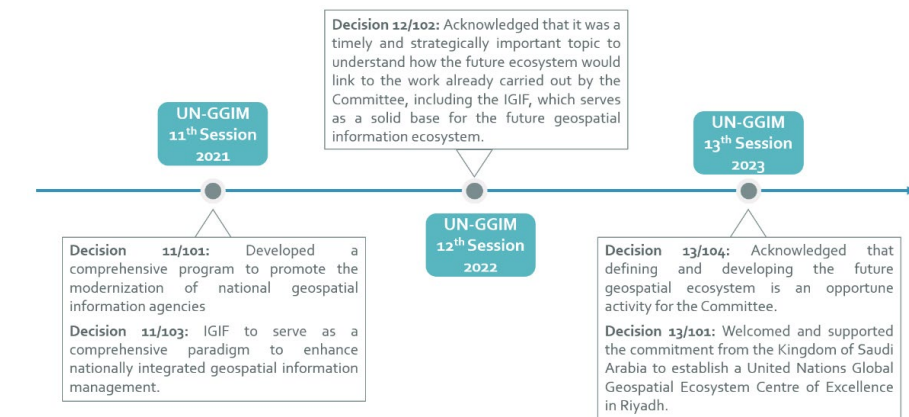


Figure 2: UN-GGIM Resolutions on Geospatial Ecosystem

These UN-GGIM discussion papers provide initial contextual understanding to deliberate on policies, standards, and initiatives that contribute to the advancement of geospatial technologies worldwide. The discussions within these UN-GGIM sessions often revolve around fostering interoperability, promoting data sharing, and establishing frameworks that support the evolution of robust geospatial ecosystems. Resolutions adopted during these sessions, summarized in more detail below, reflect the commitment of the international community to harness the full potential of geospatial information for sustainable development, disaster management, and various other critical domains.



During the Eleventh session, the Committee of Experts on Global Geospatial Information Management, which was held on August 23, 24, and 27, 2021, the decision 11/101 related to “Strengthening of geospatial information management” noted that the Committee of Experts is actively exploring avenues to enhance the global geospatial ecosystem. This includes developing a comprehensive program to promote the modernization of national geospatial information agencies and addressing the increasing geospatial requirements of developing countries.

UN-GGIM Decision 11/103, pertaining to the Integrated Geospatial Information Framework (IGIF), emphasized that the IGIF serves as a comprehensive paradigm to enhance nationally integrated geospatial information management. It underscores the importance of continuous collaboration with other emerging initiatives like the Geospatial Knowledge Infrastructure (GKI), the European Union Location Framework Blueprint, and the Geospatial Ecosystem Beyond SDIs. These collaborations provide direct interlinkages with the IGIF, ultimately extending its relevance in the future geospatial information ecosystem.

During the Twelfth session of UN-GGIM Decision 12/102 decision on “Determining the Future Geospatial Information Ecosystem” emphasized UN-GGIM’s crucial role in shaping the “future geospatial ecosystem,” emphasizing the need for synergy with existing and upcoming work items. The goal was to fortify the connections between UN-GGIM’s ongoing efforts and future initiatives, positioning the Committee of Experts strategically for the evolving geospatial landscape. In this discussion it was also emphasized the need to comprehend the future geospatial information ecosystem while recognizing the importance of sustaining the impact and continuity of the IGIF. This vision is intended to be operationalized through Country-level Action Plans.

In the discussion paper of the twelfth session, a high-level conceptual guidance was formulated, leveraging the IGIF as a practical foundation for countries as they initiate their journey toward digital transformation. The utilization of the IGIF ensures that countries can seamlessly uphold the impact and continuity of change through their IGIF Country-level Action Plans. The decision 12/102 acknowledged the timely importance of exploring the geospatial landscape amidst technological advancements, emphasizing the need to bridge the digital divide and prioritize developing countries.

The UN-GGIM Committee of Experts in its thirteenth session, decision 13/104, regarding “The Future Geospatial Information Ecosystem,” acknowledges that defining and developing the future geospatial ecosystem is an opportune activity for the Committee. However, it highlights the need for further scoping and consensus to identify and describe the foundations of the future geospatial ecosystem within the Committee’s purview.



United Nations Global Geospatial Ecosystem Centre of Excellence, Riyadh, Kingdom of Saudi Arabia

At its Tenth Plenary Meeting, the regional committee of UN-GGIM: Arab States recommended that the Kingdom of Saudi Arabia establish a United Nations Centre of Excellence in global geospatial information management in Riyadh. Subsequently, on June 20, 2023, the formal offer to host and establish a United Nations Global Geospatial Ecosystem Centre of Excellence in Riyadh was received by the UN-GGIM Secretariat from the government of the Kingdom of Saudi Arabia through its General Authority for Survey and Geospatial Information (GEOSA).

The proposed Centre aims to be sustainable, active, and internationally relevant, contributing to knowledge sharing, capacity development, and strengthening, particularly in the Arab States. GEOSA envisions the Centre as a hub for global geospatial collaboration, leveraging high-quality information and impactful analytics to address global challenges. At the thirteenth session of UN-GGIM, Decision 13/101, «Enhancing Global Geospatial Information Management Arrangements ,» the Committee of Experts welcomed and supported the commitment from the Kingdom of Saudi Arabia to establish a United Nations Global Geospatial Ecosystem Centre of Excellence in Riyadh. It encourages collaboration to clarify and refine the center’s scope and remit, ensuring alignment with the Committee’s work program.

Defining Geospatial Ecosystem

The report ‘**Towards a Sustainable Geospatial Ecosystem beyond SDIs**’ by Coetzee et al., **August 2021** explains that Geospatial ecosystem functions similarly to a natural ecosystem. It involves a dynamic community of actors - individuals, organizations, and increasingly intelligent machines - interacting in their environment through Geospatial information and technologies. The ecosystem is constantly in flux, with the balance being disrupted and restored frequently. Various stakeholders shape and coordinate the ecosystem and self-organize through competition and collaboration. Actors form temporary or long-term partnerships for their benefit or profit. The diversity of actors is crucial to the ecosystem’s success, as those who don’t add value competitively exclude each other. Some keystone actors, such as national mapping agencies or big tech companies, disproportionately impact the ecosystem.



A Geospatial ecosystem is where people from all over the world interact with each other directly or indirectly, utilizing dependable and accurate location-based data and advanced geo-analytics conveyed through dynamic geomeia or open computing environments. Like a natural ecosystem, Geospatial ecosystems are marked by solid interrelationships between actors and their surroundings, constant adaptation, and the potential for significant transformations resulting from certain events.

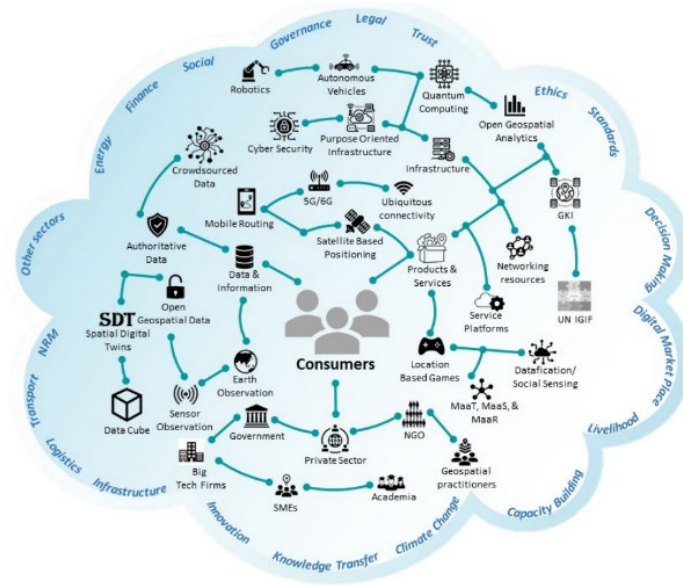


Figure 3: Geospatial Ecosystem

Source: Towards a sustainable Geospatial ecosystem beyond SDIs

Compared to natural ecosystems, information exchange doesn't deplete the source. However, the information goes through a life cycle from creation to eventual archiving or destruction. The Geospatial ecosystem provides goods and services, such as navigation for autonomous vehicles or location-based services. These goods can be tangible, such as a single dataset, or intangible, such as the results of geoprocessing events.

The Geospatial Ecosystem consists of various services and domain-specific solutions that require new approaches to education. Collaboration among stakeholders will be crucial in this regard. In this new ecosystem, up-to-date data licensing and technology platforms will be the norm. Open data will drive innovation, while web APIs will enable enhanced use of data and technology.

Services that prioritize user needs and are partly driven by AI/ML will be the focus, with software and systems providing predictive and prescriptive analytics. The presence of sophisticated and interconnected e-governance systems and procedures will further enhance the ecosystem. There will also be specialized services for scientists, experts, decision-makers, politicians, and the general population to cater to their specific needs.

Common Geospatial Ecosystem elements (Figure 4) and emerging business, technology, and emerging conditions (Figure 5) are influencing and driving the evolution and growing need for Geospatial Ecosystems that move beyond traditional spatial data infrastructures.



GEOSPATIAL ECOSYSTEM ELEMENTS

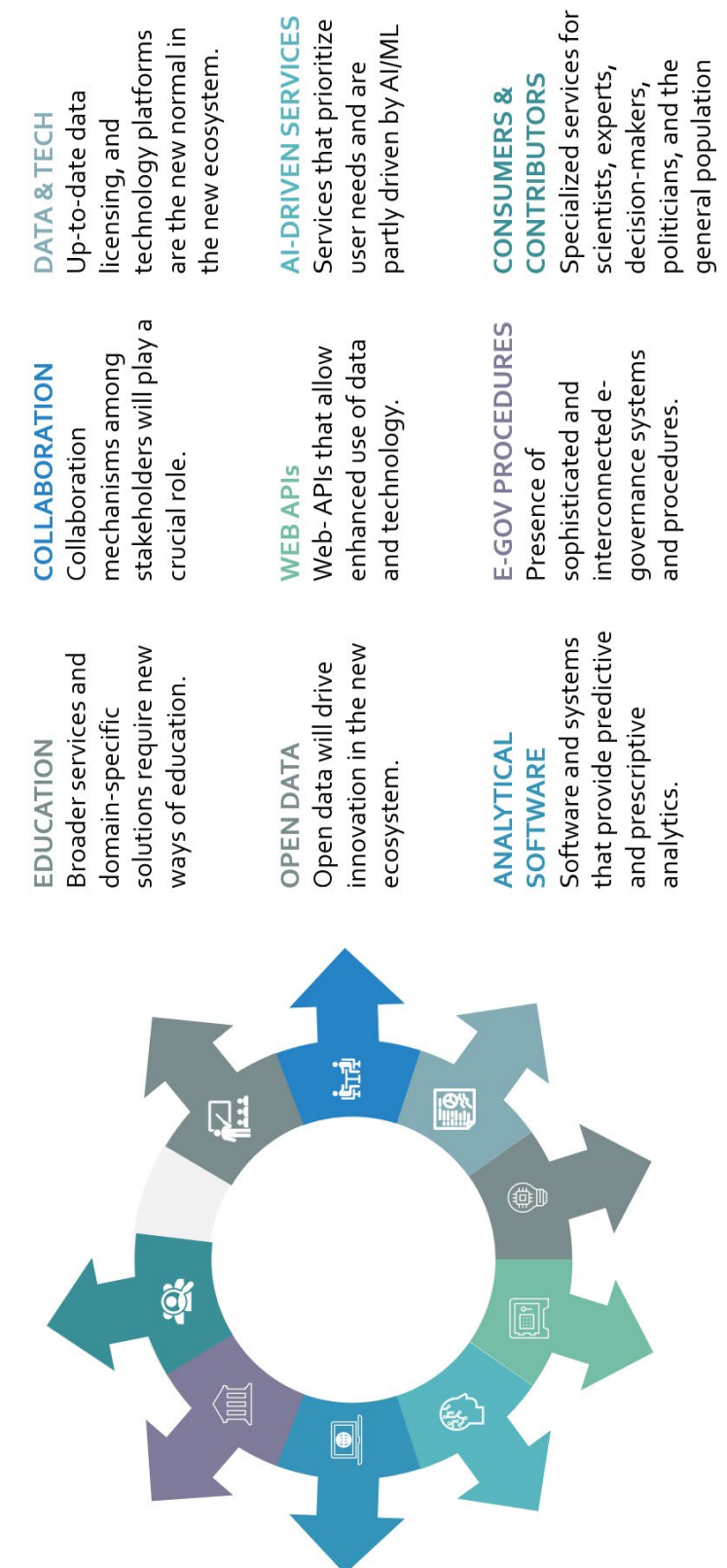


Figure 4: Geospatial Ecosystem Elements

Image source: Towards a sustainable Geospatial ecosystem beyond SDIs by Coetzee et al., August 2021 Illustration: Author's; updated.



EMERGING CONDITIONS & THEIR INFLUENCE ON THE GEOSPATIAL ECOSYSTEM



Key Features of Geospatial Ecosystem

In the future, the Geospatial ecosystem aims to be a fast and integrated system that can effectively respond to any request or emergency. This emerging ecosystem allows all sectors of society, including businesses and civil society actors, to participate fully in transformational change by bridging the digital divide with Geospatial information and analytical capabilities. This will aid in policy formulation, decision-making, and innovation, providing knowledge and shared understanding to ensure socio-economic prosperity, environmental integrity, peace, justice, national development, and a thriving global-to-local information economy.

Some of the key features of the future Geospatial Ecosystem are shown in Figure 6.

A dynamic, agile, and integrative ecosystem that caters to the needs of everyone is the future of Geospatial information management. The Kingdom of Saudi Arabia has taken the initiative to implement the first Geospatial Ecosystem that embodies all the essential features. This Geospatial Ecosystem will contribute to the national vision and international commitments. Its purpose is to ensure that no one is left behind.

Figure 5: Emerging Conditions Impacting Geospatial Ecosystem

Image source: Towards a sustainable Geospatial ecosystem beyond SDIs by Coetzee et al., 2021 Illustration: Author's; updated.



DATA MANAGEMENT

Technical standards and community norms must be established bottom-up for a seamless exchange of Geospatial info. The goal is to make public Geospatial info accessible without barriers. Accessing geospatial information in the future should be easier with a harmonized search. New methods are needed to assess data quality and trustworthiness.

AGILE ORGANIZATIONS

Governments can remain relevant by providing authoritative data, shaping regulatory environments, and facilitating the public interest. National governments can help promote the adoption of geospatial technologies through partnerships, while organizations of any size can participate in the geospatial ecosystem. Open standards are crucial to maintaining a power balance and ensuring that geospatial information remains accessible.

SKILLED WORKFORCE

As the ecosystem becomes more complex, there will be a need for Geospatial practitioners who specialize in narrow areas of expertise and generalists who have a broad understanding of various application domains within the IT field. Machine-to-machine communication and autonomous machine-centered processing and decision-making will become increasingly common.



MULTIPLE STAKEHOLDERS

The Geospatial Ecosystem will expand beyond government data providers and expert users to include various organizations, individuals, and machines in the coming years. This will involve collaboration with private sector data and service providers, academics, citizens, and NGOs, who will contribute by enhancing information and technologies through research.

SELF ORGANIZED

All actors play a crucial role in shaping the Geospatial ecosystem. Sharing essential aspects will be done in a distributed manner. Central authorities must be reviewed to ensure flexibility while adhering to standards. Maintaining the ecosystem's openness, balance, and integrity is a collective responsibility.

SOCIO-TECHNICAL SYSTEMS

The Geospatial ecosystem is a network of individuals, machines, and technology that provide location-based information and services. It is primarily decentralized and adaptable, with some closed areas that require subscription fees. The trend is shifting towards more open, loosely connected systems driven by demand. This system is flexible, redundant, and can be modified to meet diverse needs.

Geospatial Ecosystem's Aligning with UN-IGIF and SDGs

Adoption of IGIF by UN-GGIM

At its eighth session, held in New York from 1 to 3 August 2018, The United Nations Committee of Experts on Global Geospatial Information Management adopted Decision 8/113 regarding an Integrated Geospatial Information Framework (IGIF) as an overarching strategic policy guide for countries to reference when developing and strengthening their national and subnational geospatial information management systems and capabilities.

The framework was formulated by the World Bank, with contributions from the many actors within the Committee of Experts as an important and practical global guide. The Committee of Experts supported the approach that the framework should serve as a living document to be further refined as technologies, processes, and knowledge evolve.

The Committee of Experts acknowledged that the integrated geospatial information framework was designed to provide a comprehensive guideline for all of the Committee's activities. Therefore, they agreed that the implementation guide should reference and build upon the current activities, guidelines, and frameworks of the Committee's working groups as well as address national and regional spatial data infrastructure and related initiatives and frameworks. This is particularly important in the context of each of the nine strategic pathways of the framework.

At its ninth session, held in New York from 7 to 9 August 2019, The United Nations Committee of Experts on Global Geospatial Information Management adopted Decision 9/103 regarding Integrated Geospatial Information Framework, as a basis and guide for developing, integrating and strengthening geospatial information management, and to assist countries in bridging the geospatial digital divide, according to national priorities.

The UN-GGIM Secretariat undertook a series of extensive and productive expert consultative meetings that had been convened to develop and refine the Implementation Guide in preparation for subsequent and broader global consultations with Member States and other key stakeholders.

It was also suggested that when it came to implementing the Integrated Geospatial Information Framework, it was necessary to build on existing regional experiences in the implementation of national spatial data infrastructure and to recognize that there was not only one way to implement the Framework.



Figure 6: Features of Geospatial Ecosystem
Image source: Towards a sustainable Geospatial ecosystem beyond SDIs by Coetzee et al., 2021 Illustration: Author's; updated.



IGIF Overview

According to the United Nations – Integrated Information Framework (IGIF) part 1 Overarching Strategy, the IGIF is a multi-dimensional framework that guides countries in the development, management, and refinement of their national geospatial information resources, anchored by and implemented through nine (9) strategic pathways in three (3) main areas of influence: governance, technology, and people (Figure 7).

IGIF Vision

The efficient use of geospatial information by all countries to effectively measure, monitor, and achieve sustainable social, economic, and environmental development - leaving no one behind.

IGIF Mission

To promote and support innovation and provide the leadership, coordination, and standards necessary to deliver integrated geospatial information that can be leveraged to find sustainable solutions for social, economic, and environmental development.

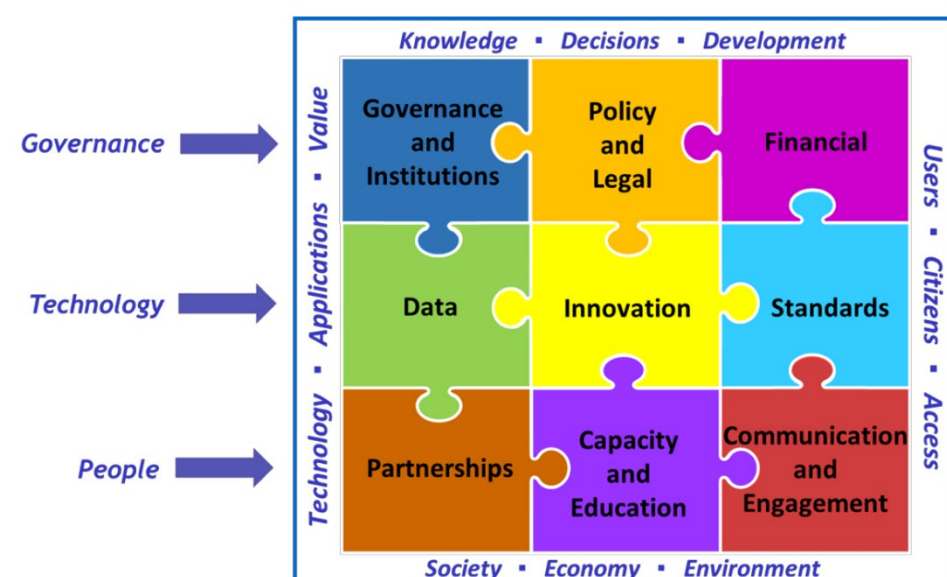


Figure 7: Nine Strategic Pathways of the UN-IGIF
Source: United Nations – Integrated Geospatial Information Framework part 1 Overarching Strategy

Although initially developed to provide a basis and guide for lower to middle-income countries to reference when developing and strengthening their national and sub-national arrangements in geospatial information management and related infrastructures, the UN-IGIF has evolved as a living document. It has become apparent that many high-income and developed countries are also significantly benefitting from the integrative and inclusive strategic nature of the Framework.

The United Nations Integrated Geospatial Information Framework (UN-IGIF) can be used to inform and contribute to national development plans. The Framework highlights how relevant geospatial information can be utilized while exploiting accessible and available technologies to support developing countries and regions to establish and enhance geospatial capabilities

through the use of appropriate frameworks, methods, guidelines, and standards which can be piloted, replicated, and delivered within and across countries and regions.

The UN-IGIF presents a forward-looking approach that creates an enabling environment where national governments can coordinate, develop, strengthen, and promote the efficient and effective use and sharing of geospatial information for policy formulation, decision-making, and innovation. It establishes a common vision for all government agencies and expresses the goals that will realize the vision, the actions that need to be implemented to achieve the goals, and the benefits necessary to support national development.

Importantly, the UN-IGIF is not an infrastructure. It is a standalone 'framework,' independent of Spatial Data Infrastructures (SDIs), National Spatial Data Infrastructures (NSDIs), and any other infrastructures. However, the UN-IGIF fundamentally recognizes, builds upon, and augments previous investments and substantial achievements in planning and implementing SDIs and NSDIs.

The UN-IGIF also provides a mechanism by which citizens and the community can discover, view, and obtain meaningful and accurate information about their country and community from different organizations while reducing the burden on the user to locate, access, and use traditionally disintegrated data themes. Community participation is an integral part of the UN-IGIF. Local knowledge, in conjunction with scientific methods and government data resources, enhances our understanding of our natural and built environments.

The UN-IGIF identifies eight (8) aspirational goals and seven (7) underpinning principles that represent the key characteristics and values that are to be used by countries as a guide when implementing the Framework. How these principles are applied will depend on the implementation approach adopted by each country. Adherence to these principles will deliver consistent geospatial information management, resulting in more open, accountable, responsive, and efficient government. to achieve the overarching vision.

IGIF Aspirational Goals

- ✓ 1: Effective Geospatial Information Management
- ✓ 2: Increased Capacity, Capability, and Knowledge Transfer
- ✓ 3: Integrated Geospatial Information Systems and Services
- ✓ 4: Economic Return on Investment
- ✓ 5: Sustainable Education and Training Programs
- ✓ 6: International Cooperation and Partnerships
- ✓ 7: Enhanced National Engagement and Communication
- ✓ 8: Enriched Societal Value and Benefits

IFIG Underpinning Principles

- ✓ 1: Strategic Enablement
- ✓ 2: Transparent and Accountable
- ✓ 3: Reliable, Accessible and Easily Used
- ✓ 4: Collaboration and Cooperation
- ✓ 5: Integrative Solution
- ✓ 6: Sustainable and Valued
- ✓ 7: Leadership and Commitment



IGIF Strategic Pathways

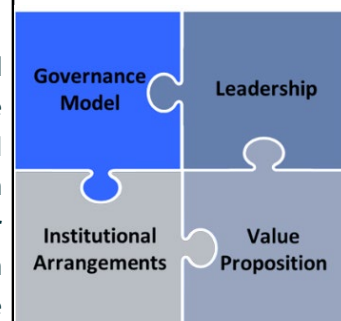
The UN-IGIF is anchored by and implemented through nine (9) strategic pathways in three (3) main areas of influence: governance, technology, and people. These strategic pathways aim to guide governments toward implementing integrated geospatial information systems in a way that will deliver a vision for sustainable social, economic, and environmental development.

The strategic pathways are presented as separate pieces of a jigsaw puzzle in recognition that there are many aspects and dimensions to each individual pathway and that when joined together, the UN-IGIF is connected, integrated, and implemented. Figure 7 illustrates the nine strategic pathways surrounded by some benefits that can be realized when implemented together.

Each strategic pathway contains four key elements as the enabling components or requirements for the strategic pathway to be implemented. The key elements are a derivative of the pathways at a lower level of detail and are intended to enable concepts to be more concrete and understandable for countries.

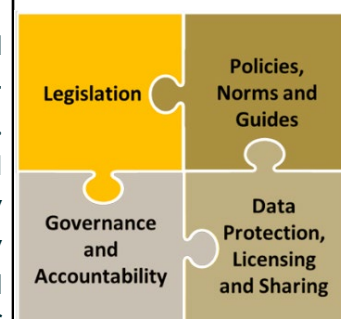
STRATEGIC PATHWAY 1: Governance and Institutions

This strategic pathway aims to establish a leadership and governance model, institutional arrangements, and a clear value proposition to strengthen multi-disciplinary and multisectoral participation in achieving the UN-IGIF. The objective is to gain political support, strengthen institutional mandates, and foster a cooperative data-sharing environment through a shared vision and understanding of the value of the UN-IGIF, as well as the roles and responsibilities required to achieve this vision.



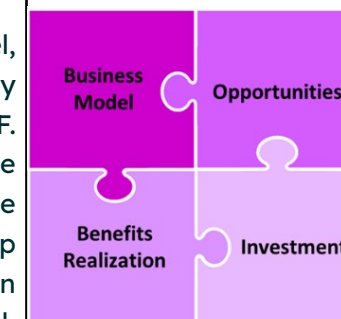
STRATEGIC PATHWAY 2: Policy and Legal

Ensuring effective, efficient, and secure management and exchange of geospatial information at both national and sub-national levels requires a solid policy and legal framework. The aim is to improve policies and laws that impact geospatial information management by proactively monitoring the policy and legal environment. This includes assigning responsibility for data production and staying up to date with issues and challenges related to the innovative and creative use of geospatial information and emerging technologies.



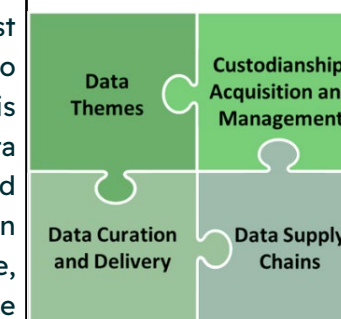
STRATEGIC PATHWAY 3: Financial

This strategic pathway determines the business model, establishes financial partnerships, and identifies the necessary investments and financing means for delivering UN-IGIF. The plan also recognizes the milestones required to achieve and maintain momentum to realize benefits. The objective is to understand the financial framework required to set up and sustain UN-IGIF, as well as a long-term investment plan that enables the government to respond to evolving societal, environmental, and economic demands for geospatial data.



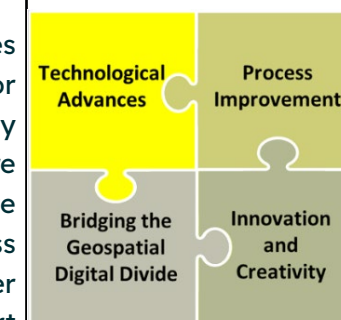
STRATEGIC PATHWAY 4: Data

This strategic pathway establishes guidelines for the best practices of collecting and managing geospatial data, while also defining the framework for its custodianship. The objective is to enable data custodians to meet their obligations for data management, sharing, and reuse with the government and the user community. This is achieved through the execution of well-defined data supply chains that organize, plan, acquire, integrate, manage, maintain, curate, publish, and archive geospatial information, which is appropriate for establishing and sustaining cross-sector and multidisciplinary collaboration.



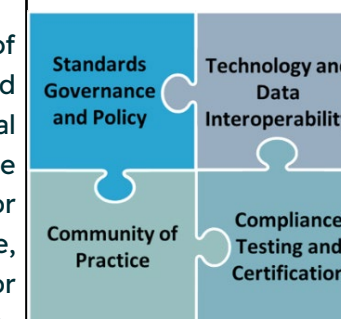
STRATEGIC PATHWAY 5: Innovation

This strategic pathway recognizes that technology and processes are continuously evolving, creating enhanced opportunities for innovation and creativity that enable governments to quickly bridge the digital divide and transition towards the future geospatial information ecosystem. The objective is to stimulate the use of the latest cost-effective technologies, process improvements, and innovations so that governments, no matter what their current situation is, may leapfrog to state-of-the-art geospatial information management systems and practices.



STRATEGIC PATHWAY 6: Standards

This strategic pathway establishes and ensures the adoption of standards and compliance mechanisms for enabling data and technology interoperability to deliver integrated geospatial information and to create location-based knowledge. The objective is to enable an efficient and consistent approach for different information systems to be able to discover, manage, communicate, exchange, and apply geospatial information for a multitude of uses, improving understanding, and decision-making.





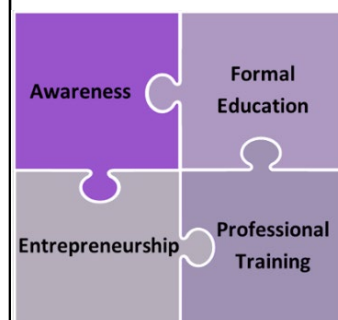
STRATEGIC PATHWAY 7: Partnerships

This strategic pathway establishes cross-sector and interdisciplinary collaboration, cooperation, and coordination with all levels of government, the geospatial industry, the private sector, academia, and the international community as an important premise for developing and sustaining the UN-IGIF. The objective is to create and sustain the value of geospatial information through a culture based on inclusion, trusted partnerships, and strategic alliances that recognize common needs, aspirations, and goals toward achieving national priorities and outcomes.



STRATEGIC PATHWAY 8: Capacity and Education

This strategic pathway establishes enduring capacity development and education programs so that the value and benefits of integrated geospatial information management are sustained for the longer term. The objective is to raise awareness and build and strengthen knowledge, competencies, skills, instincts, processes, resources, and innovative entrepreneurship that organizations, communities, and individuals require to utilize geospatial information for evidence-based decision-making and effective service delivery.



STRATEGIC PATHWAY 9: Communication and Engagement

This strategic pathway recognizes that stakeholder identification, user engagement, and strategic communication are essential to successfully deliver UN-IGIF arrangements nationally and sub-nationally for sustainable social, economic, and environmental development. The objective is to ensure effective communication and engagement to enhance and deepen participation and contributions from all stakeholders at all levels. Commitment, mutual understanding, collaboration, cooperation, and communication are essential to implementing the UN-IGIF with organizations and stakeholders successfully.



In summary, the United Nations Integrated Geospatial Information Framework (UN-IGIF) unfolds as a guiding beacon for the Geospatial Ecosystem, offering strategic pathways in three main areas of influence: governance, technology, and people.

In terms of **governance**, the IGIF establishes principles and frameworks that ensure effective and harmonized geospatial data management. By adhering to these governance standards, the Geospatial Ecosystem can develop trust, facilitate collaboration, and address socio-economic challenges related to data sharing.

In the realm of **technology**, the UN-IGIF sets the stage for the adoption of cutting-edge geospatial tools and methodologies. This ensures that the Geospatial Ecosystem remains

technologically relevant, adaptive to emerging trends, and capable of leveraging advancements for sustainable development.

The emphasis on **people** underscores the significance of capacity building, education, and inclusivity. By prioritizing these aspects, the Geospatial Ecosystem can nurture a skilled and diverse community of practitioners, fostering innovation and ensuring that the benefits of geospatial information are accessible to all.

In essence, the UN-IGIF's strategic pathways in governance, technology, and people collectively contribute to the resilience, effectiveness, and national and global impact of the Geospatial Ecosystem.

Geospatial Ecosystem supporting IGIF Pathways

The Geospatial Ecosystem plays a crucial role in advancing the objectives of the United Nations Integrated Geospatial Information Framework (UN-IGIF) by facilitating the broader utilization of geospatial information and its seamless integration with various other datasets and systems, providing a tangible and dynamic implementation of key IGIF pathways.

As outlined in the background paper, “Towards a sustainable geospatial ecosystem beyond SDIs” by Coetzee et al, 2021 the emerging conditions are enabling the Geospatial community to transition to a sustainable Geospatial Ecosystem. Each of the identified characteristics of the Geospatial Ecosystem supports one or more of the IGIF Strategic Pathways. This intrinsic link strengthens the overall governance, technological advancements, and people-centric approach to Geospatial information management within the country.

From encouraging for increased coordination to fostering active collaborations and trans-disciplinary research, these characteristics form a strategic guide for the Geospatial Ecosystem's development, emphasizing adaptability, collaboration, and lifelong learning. The paper also emphasizes the importance of these characteristics in shaping a dynamic, collaborative, and well-coordinated Geospatial Ecosystem capable of meeting the evolving demands of governance, society, industry, and government.

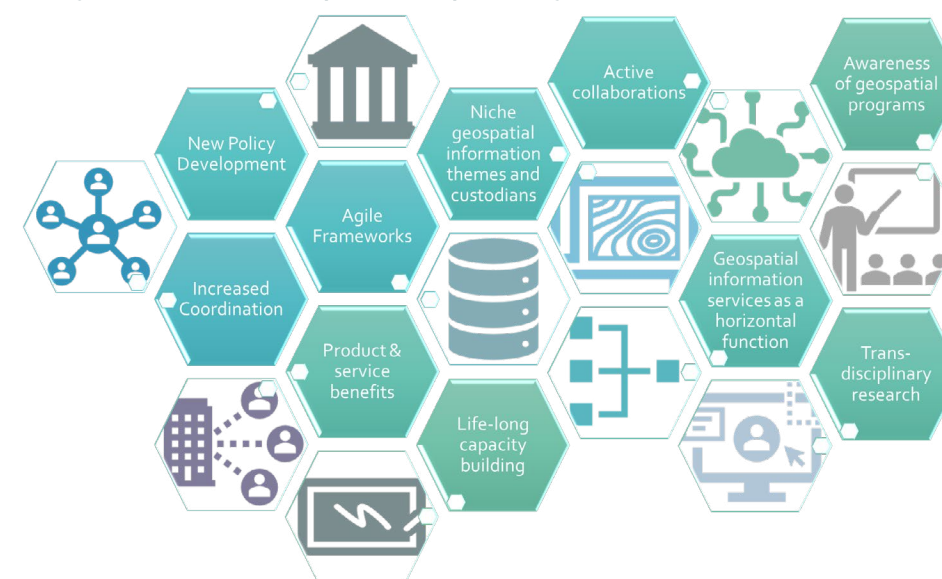


Figure 8: Geospatial Ecosystem Characteristics



The characteristics of the Geospatial Ecosystem discussed below are strongly relevant to the strategic pathways delineated by the United Nations Integrated Geospatial Information Framework.

1. **Increased coordination:** The Geospatial Ecosystem demands more collaboration and lesser control by the lead government organizations while at the same time continuing in a coordinating capacity, for example, on the acquisition of aerial or other imagery, to avoid duplication. The emphasis should not be to control what is done (other than in clearly defined areas) but to recognize, understand, and derive advantages from understanding the diversity of approaches, data, processes, analytics, and output display forms used. An overriding aim should be to ensure that new project-based silos do not emerge again and that optimum opportunities are created to share information.

2. **Agile frameworks:** In the Geospatial Ecosystem approach, governments will work on developing flexible frameworks or guiding principles that enable different initiatives or projects to be interoperable and meet the specific needs of users. This applies to both initiatives or projects undertaken by the public sector and those funded by consultants. In the case of consultant-funded projects, public sector bodies would require consultants to provide all relevant data and details about the processes and analytics used in their contracts of appointment.

3. **New policy development:** Developing and maintaining an agile framework requires a comprehensive set of policies, such as those related to pricing, licensing, use and reuse, and standards. In the ecosystem environment, governments must incentivize the custodians (or the representatives of custodian organizations) with rewards that are related to their performance criteria or cost savings to ensure the adoption of the new policies.

4. **Niche geospatial information themes and custodians:** Organizations within the Geospatial Ecosystem responsible for geospatial data would continue to identify important themes and custodians for those themes. However, the focus would be on gathering information that is crucial for governance and society's functioning, which is not easily accessible elsewhere. The government should not attempt to impede or prohibit alternative data sources for other areas and themes. Collaboration is key in this regard. For example, there may be a custodian for municipal data and a coordinating custodian for the national dataset.

5. **Active collaborations:** Small public sector bodies can enhance their competitive advantage, survive, and thrive in an ecosystem by collaborating with others, where collaborators could be interpreted broadly. For instance, OpenStreetMap and open-source software developers, partnerships with private companies to maintain their datasets, the IT support division in their organization, or an NGO could be considered potential collaborators for smaller municipalities. This collaboration should focus on gathering information essential for the municipality's current functioning instead of creating a data warehouse for potential future use.

6. **Geospatial information services as a horizontal function:** In larger public sector organizations, it will be better to provide or share geospatial information support as a horizontal



function, rather than having it seated in a single division like planning or civil engineering. It would be beneficial to have a single entity that is well-funded and has a clearly defined cross-administration support role. This entity could facilitate automation of data and analytics sharing, problem-solving, opportunity seeking, and sharing solutions found in one department or agency with other agencies, as well as updates on new technical, guidance, and legislative developments.

7. **Product and service benefits through implementations:** Industry stakeholders should demonstrate key capabilities operating on basic evolving infrastructure such as the web. In an ecosystem shaped by competition, the viability and applicability of industry vendor solutions will be judged by testing (at least) and assessing operative pilot implementations.

8. **Trans-disciplinary research:** Universities' computer science and IT programs are crucial for advancing key technologies such as information search, AI-based filtering and suggestions, and data fusion. Geospatial experts possess knowledge of the subject matter, but not necessarily of data or computation. Therefore, researchers in the geospatial field should actively collaborate and form partnerships with their colleagues in the computer science and IT fields. Additionally, trans-disciplinary research is essential for both existing and new application domains to ensure the optimal usage of geospatial information and technologies. It is also necessary to collaborate with business schools and other stakeholders working on the economic models of such large cooperative ecosystems. Lastly, trans-disciplinary research is required to examine the social impacts of the widespread use of geospatial information and technologies.

9. **Awareness of geospatial programs and their importance for society, industry, and government:** To meet the current and future demands for skills in the geospatial ecosystem, there will need to be a significant increase in the number of graduates in study programs that focus specifically on geospatial information and technologies. To achieve this, awareness must be raised of such programs and the importance of geospatial information and analytics for society, industry, and government. Educators from all over the world should intensify their collaboration to achieve this goal. One way to do this is by marketing their offerings under a common, all-inclusive name instead of the current diversity of unique labels, such as geomatics, geomedia, geomarketing, geoinformatics, and geographic information science (GISc or GIScience). Both the public and private sectors need to collaborate and support this effort.

10. **Life-long capacity building:** It is important to develop skills and provide opportunities for mid-career professionals who received their education and training before the current ecosystem was established. This should not be an option but rather a regular requirement going forward. In addition to changes in university curricula aimed at undergraduate and graduate students, there is a need to focus on the development of mid-career professionals. It is not practical for propeller-certified pilots to fly modern jets. Therefore, it is necessary to update the skills of mid-career professionals to ensure that they are well equipped to serve the ecosystem effectively.

The characteristics of the Geospatial Ecosystem, as articulated above, strongly align with



the strategic pathways delineated by the United Nations Integrated Geospatial Information Framework. This synergistic relationship is appropriately illustrated in Table 1, underlining how the Geospatial Ecosystem’s characteristics and UN-IGIF pathways converge to build a robust and sustainable approach to geospatial information management.

Geospatial Ecosystem Characteristics	IGIF Strategic Pathways	SP1 - Governance and Institutions	SP2 – Policy & Legal	SP3 - Financial	SP4 - Data	SP5 - Innovation	SP6 - Standards	SP7 - Partnerships	SP8 – Capacity & Education	SP9 – Comms & Engagement
	Increased coordination	✓				✓	✓	✓	✓	✓
	Agile frameworks	✓	✓			✓		✓		✓
	New policy development	✓	✓			✓	✓		✓	
	Niche geospatial information themes	✓	✓	✓	✓			✓		✓
	Active collaborations	✓		✓		✓		✓	✓	
	Geospatial as a horizontal function	✓	✓		✓		✓	✓	✓	✓
	Product and service benefits through implementations			✓	✓	✓	✓	✓		
	Trans-disciplinary research				✓	✓	✓	✓	✓	
	Awareness of geospatial programs and their importance	✓						✓	✓	✓
	Life-long capacity building					✓		✓	✓	✓

Table 1: Geospatial Ecosystem characteristics aligning to United Nations Integrated Geospatial Information Strategic Pathways



Geospatial Ecosystem working for the Sustainable Development Goals

The Sustainable Development Goals (SDGs) provide a roadmap to achieve a better and more sustainable future for everyone. They tackle the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace, and justice. The 17 SDGs are shown in Figure 9 below. These 17 goals are interrelated, and it is crucial to achieve them all by 2030 to ensure that no one is left behind.



Figure 9: Sustainable Development Goals

Governments rely heavily on high-quality, timely, reliable, and disaggregated data to measure, monitor, and achieve the 17 Sustainable Development Goals. The document Transforming our World: The 2030 Agenda for Sustainable Development clearly states the need for the availability and use of such data, including earth observation and geospatial information for supporting and tracking progress.

As already discussed in the previous section, the UN-GGIM framework on Integrated Geospatial Information (UN-IGIF) supports countries to take practical measures to implement digital transformation and bridge the spatial digital divide in implementing national strategic priorities and the 2030 Agenda for Sustainable Development. UN-IGIF and its guidelines build on the existing work of the United Nations Committee of Experts on Global Geoinformation Management (UN-GGIM) and now provide the basis for the Committee’s work program to support the implementation of the 2030 Agenda for Sustainable Development and other goals. The Economic and Social Council (ECOSOC), the umbrella body of UN-GGIM, recognized and reiterated the importance of strengthening and increasing the effectiveness of UN-GGIM,



especially for achieving its contribution to the Sustainable Development Goals and SDG responses.

Figure 10 illustrates the relation of the Integrated Geospatial Information Framework required to achieve the SDGs. Progress under the 17 Sustainable Development Goals (SDGs) are measured through 169 targets and 231 indicators that rely on various data sources, including satellite imagery, population and demography data, cell phone data, topography, and more.

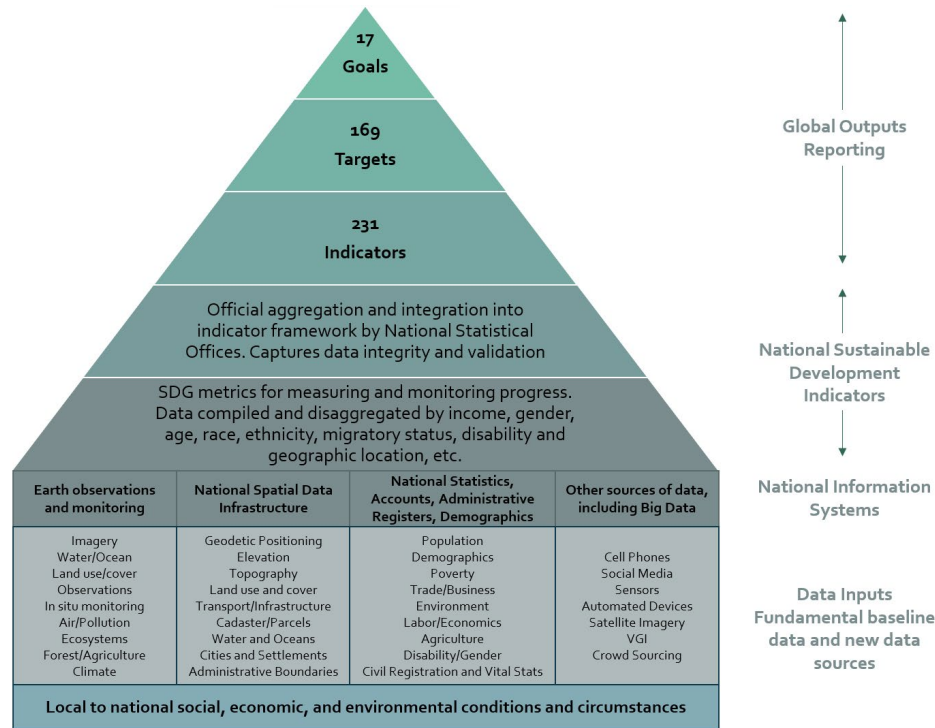


Figure 10: Integrated Geospatial Information required for achieving Sustainable Development Goals
Source: UN-GGIM Secretariat

All of these data sources are fed into national information systems, like earth observations and monitoring systems, national spatial data infrastructure, national statistics, registers, etc. that are used to compile disaggregated data on factors such as income, gender, age, race, ethnicity, migratory status, disability, and geographic location. This disaggregated data is then used at the national level to create official indicators that measure and monitor progress towards the SDGs, with the National Statistical Offices responsible for aggregating and integrating the data.

According to decision 7/104, UN-GGIM adopted a list of 14 basic global geospatial themes at its seventh session. The implementation of these topics requires the integration of information from various agencies, including the National Spatial Data and Mapping Agency and the National Bureau of Statistics, to produce standardized basic data. The data will support the implementation of the 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals (SDGs). They are also an important foundation for a comprehensive geospatial information framework and a core output from the Geospatial Ecosystem.

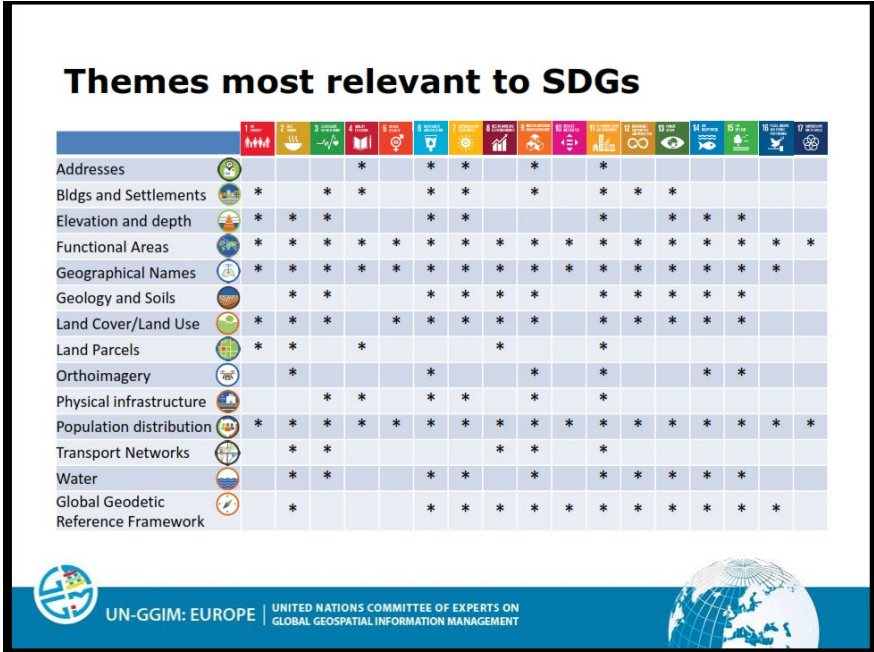


Figure 11: Global Fundamental Geospatial Data Themes most relevant to the SDGs
Source: UN-GGIM Secretariat

In the contemporary interconnected digital landscape, the Geospatial Ecosystem contributes significantly to the realization of Sustainable Development Goals (SDGs) through several key mechanisms.

One essential aspect is the emphasis on Partnerships and Collaborations. The Geospatial Ecosystem promotes collaborative efforts among stakeholders, fostering partnerships that enhance the effectiveness of geospatial information in addressing the challenges posed by the SDGs. These partnerships not only leverage diverse expertise but also encourage sharing of resources and knowledge, creating a synergistic approach to sustainable development.

Agile Organizations represent another facet of the Geospatial Ecosystem's support for UN-IGIF. The adaptability and responsiveness of organizations to changing circumstances are crucial for successfully navigating the complexities associated with SDGs. The Geospatial Ecosystem facilitates the development of agile organizations that can swiftly adjust strategies and operations to align with evolving geospatial and informational requirements.

Furthermore, the Availability of Integrated Datasets is a key enabler. The Geospatial Ecosystem ensures that relevant datasets are seamlessly integrated, providing a comprehensive understanding of geospatial factors in the context of the SDGs. This integration enhances the quality and depth of information available for decision-makers, enabling them to make more informed and impactful choices.

Insights for Decision Making is another critical contribution. By harnessing geospatial information, the Geospatial Ecosystem empowers decision-makers with valuable insights. These insights go beyond raw data, offering a nuanced understanding of spatial relationships and patterns, thereby facilitating more effective and targeted decision-making processes aligned with the SDGs.

Advanced Visualizations represent a powerful tool within the Geospatial Ecosystem. By employing sophisticated visualization techniques, complex geospatial data is translated into



easily understandable representations. This not only enhances communication but also aids in conveying information to a broader audience, including those without specialized technical knowledge.

Data Contextualization and Interoperability is achieved through the establishment of a connection between semantics and natural language. This innovation makes geospatial information accessible to a wider audience. By bridging the technical and linguistic aspects, the Geospatial Ecosystem ensures that geospatial insights are comprehensible and usable across diverse user groups.

Automation is a fundamental aspect of the Geospatial Ecosystem's contribution. By making processes repeatable and minimizing human intervention, automation enhances efficiency and accuracy. Standardizing processes and promoting repeatability further ensures consistency and reliability in utilizing geospatial information for achieving the SDGs.

The Bottom-up Approach is a guiding principle within the Geospatial Ecosystem. This approach emphasizes grassroots involvement and engagement, recognizing the significance of local perspectives and community-level initiatives in addressing the SDGs. By empowering local communities, the Geospatial Ecosystem contributes to a more inclusive and sustainable approach to geospatial information utilization for global development goals.

Saudi Arabian National Geospatial Ecosystem Requirements

The Saudi Arabian National Geospatial Ecosystem aims to integrate and harmonize various components of the Geospatial ecosystem, including data, tools, standards, and policies. Following extensive engagement with government, non-government, and international stakeholders the General Authority for Survey and Geospatial Information (GEOSA) identified several critical factors outlined in Figure 13 that are required to establish a sustainable, successful, and nationally appropriate Geospatial Ecosystem in Saudi Arabia.

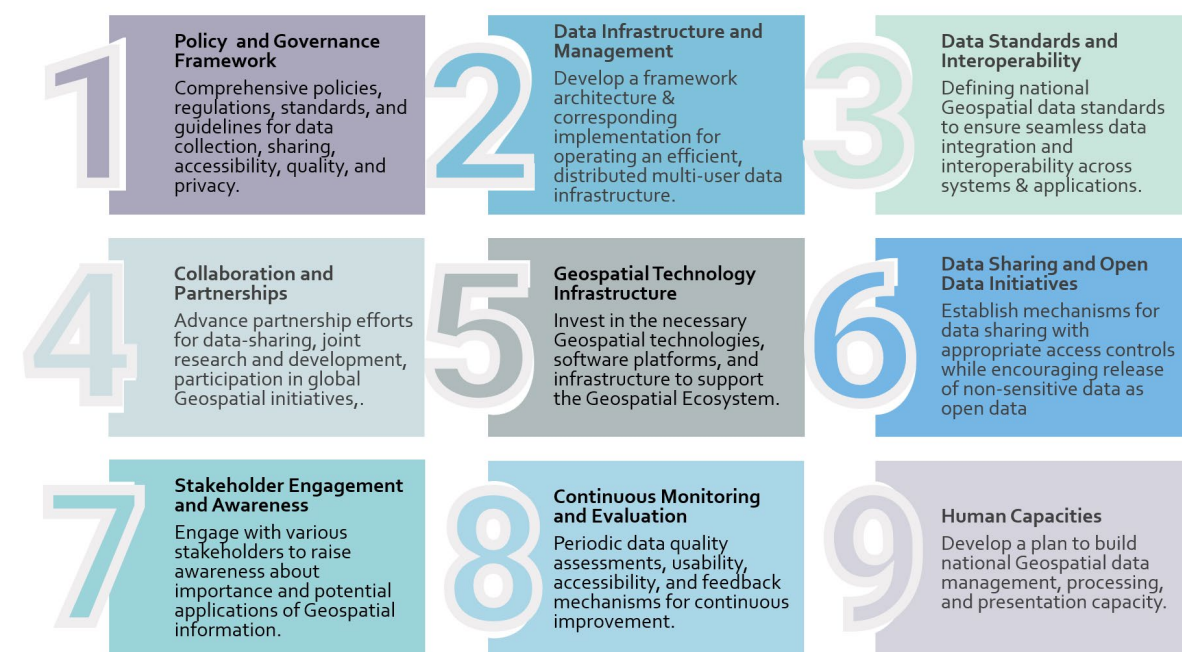


Figure 12: Factors influencing Saudi Arabian Geospatial Ecosystem



The establishment of the Saudi Arabian National Geospatial Ecosystem is expected to deliver numerous benefits to the Kingdom, including reduced data duplication, fostering interoperability, productivity enhancement, increased data sharing and ensuring data dissemination standards, to name a few.

Broadly speaking, the Saudi Arabian National Geospatial Ecosystem will constitute of five high-level components, as outlined below.

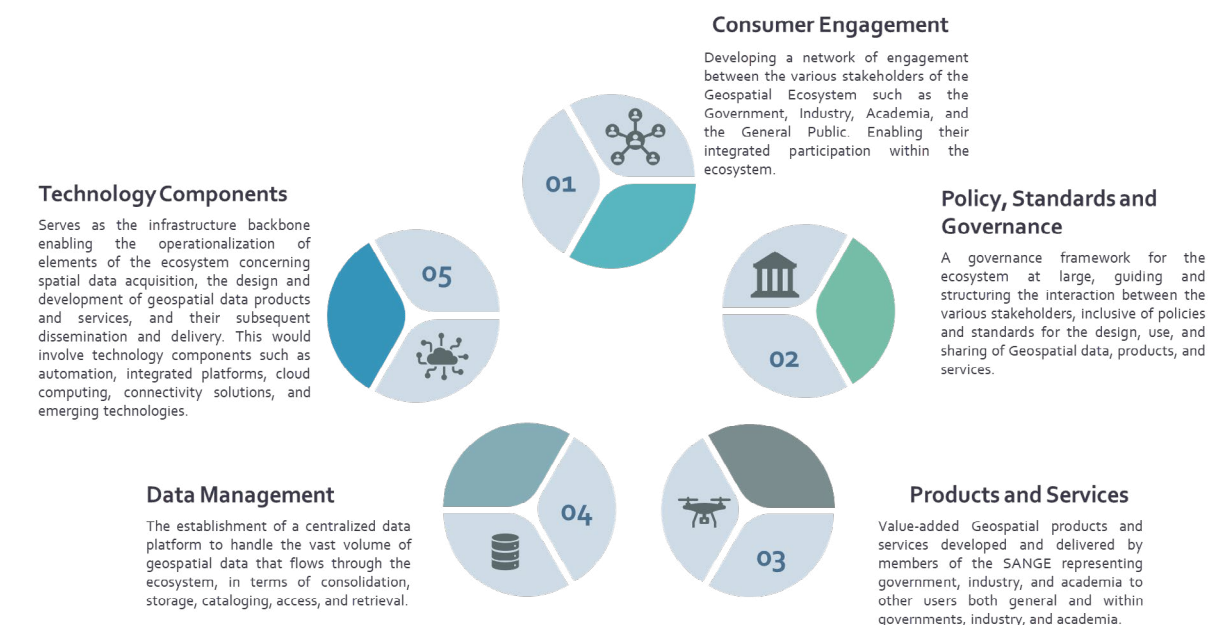


Figure 13 - Saudi Arabian National Geospatial Ecosystem high-level components

This categorization helps provide a structured approach to addressing the diverse requirements within the Geospatial ecosystem. By organizing the requirements into these components, GEOSA & its stakeholders can gain a comprehensive view of the different aspects that need to be addressed to establish and operate a successful Geospatial ecosystem.

This classification facilitates effective planning, resource allocation, and decision-making in developing and implementing the necessary strategies, policies, products, services, data management practices, and technological capabilities to support the objectives of Saudi Arabian National Geospatial Ecosystem.

Consumer Engagement

The requirements within the Consumer Engagement component underline the significance of stakeholder engagement and user-centric approaches in driving effective Geospatial initiatives, promoting improved collaboration, and delivering enhanced value to the Saudi Arabian National Geospatial Ecosystem ecosystem's users. Under the Consumer Engagement component, there are two key requirement

parameters:

- **Stakeholders Engagement:** This parameter emphasizes fostering better coordination, collaboration, and synergy among diverse groups and stakeholders within the Geospatial



ecosystem. The objective is to promote effective and impactful Geospatial initiatives by facilitating active engagement and participation of stakeholders. This can involve various activities such as regular communication, consultations, workshops, and partnerships to ensure the alignment of goals and maximize the collective benefits derived from Geospatial initiatives.

- **Value-Impact Centric (User Views):** This parameter focuses on meeting the growing expectations of modern users by providing user-friendly GIS platforms, web mapping applications, and mobile apps. The aim is to enhance user experiences and engagement by offering more interactive and engaging features. By adopting a value-impact-centric approach, the Geospatial ecosystem aims to understand and cater to the needs and preferences of users, providing them with accessible and intuitive tools for leveraging Geospatial data and services.

Policy, Standards, and Governance



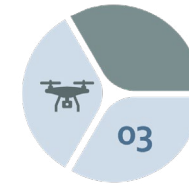
The requirements within the Policy, Standards, and Governance component emphasize the importance of establishing a strong governance framework, promoting data standards and interoperability, and ensuring proper metadata documentation. By prioritizing these requirement parameters, Saudi Arabian can establish a solid foundation for effective governance, standardized data practices, and comprehensive metadata

documentation, enabling seamless collaboration, interoperability, and reliable decision-making across the ecosystem. Under this component, there are three key requirement parameters:

- **Governance and Policy Framework:** This parameter emphasizes the need for a robust governance and policy framework to guide the development and implementation of the Saudi Arabian National Geospatial Ecosystem. It involves establishing effective governance frameworks and policies to ensure responsible data management, privacy protection, data access rights, and ethical considerations related to Geospatial data such as GKI, IGIF, OGC Geospatial standards, ISO standards, etc.
- **Data Standards and Interoperability:** This parameter focuses on the importance of standardized data formats, structures, and protocols to enable seamless data integration and interoperability within the Geospatial ecosystem. By adhering to common data standards and Geospatial data formats, such as GeoJSON, Shapefile, or GeoTIFF, different systems and applications can effectively exchange and share Geospatial data, facilitating better data integration, analysis, and decision-making. Interoperability ensures that data from various sources can be easily accessed, combined, and utilized Geospatial data effectively.
- **Metadata Definition:** This parameter highlights the need for well-defined metadata standards and practices. Metadata provides essential information about Geospatial data, such as its source, quality, and usage. By establishing robust metadata definitions and guidelines, the Geospatial Ecosystem can ensure consistent and accurate data documentation, enabling users to understand and assess its reliability and suitability for specific purposes. Proper metadata management enhances data discoverability, accessibility, and usability of Geospatial datasets.



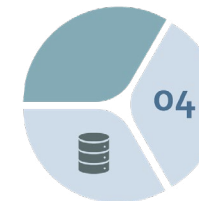
Products and Services



Under the Product and Services component, there are six key requirement parameters. By addressing these parameters, Saudi Arabian National Geospatial Ecosystem can enhance data accessibility, analysis, and capabilities, delivering valuable and innovative Geospatial products and services to meet the evolving needs of stakeholders and users.

- **Data Sharing:** This parameter enables seamless access and integration of Geospatial information for the public, irrespective of administrative barriers. It emphasizes breaking down data silos and promoting open data initiatives to facilitate broader data sharing and collaboration.
- **Metadata-based Cataloging:** This parameter emphasizes implementing robust metadata standards and cataloging mechanisms. It aims to enhance Geospatial datasets discovery, assessment, and understanding by providing comprehensive and standardized metadata information.
- **Geo-analytics Capabilities:** This parameter highlights the importance of implementing geo-analytics capabilities to unlock the full potential of Geospatial data. It involves leveraging advanced automated analytical tools and techniques to extract valuable insights, enabling informed decision-making and problem-solving.
- **Advanced Augmented Analytics:** This parameter focuses on applying advanced analytics techniques, such as machine learning and artificial intelligence, to Geospatial data. It aims to automate data exploration, generate automated insights (such as trend analysis, anomaly detection, prediction, and forecasting), and provide decision support based on sophisticated analytical models.
- **Spatial Digital Twins:** This parameter emphasizes developing and deploying use-cases by spatial digital twins. It involves creating virtual replicas of physical assets or systems, integrating Geospatial data with other data sources, and utilizing technology frameworks to enable advanced analytics, simulation, and scenario planning.
- **Positioning and Navigation:** This parameter enables use-case and end-user applications that rely on precise positioning and navigation functionalities. It involves leveraging technologies such as Global Navigation Satellite Systems (GNSS) and other positioning systems to provide accurate location information for various applications.

Data Management



The rapid development in technology leads to an enormous amount of data. The data collected needs structuring to specific functional characteristics. In this component, the desired Geospatial ecosystem focuses on seven key requirement parameters such as:

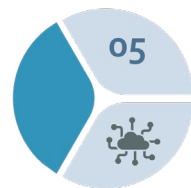
- **Geospatial-centric:** This parameter emphasizes the importance of adopting a spatial-centric approach, prioritizing geographical location in decision-making and daily operations. It uses Geospatial data as a foundation for evidence-based decision-making and operational activities.
- **Structured/Unstructured Data Integration:** This parameter integrates Geospatial data



with other data sources, such as sensor networks, IoT devices, and social media. It enables the creation of rich-content applications that provide users with relevant information based on location or user interests.

- **Data Cube:** This parameter involves developing and deploying a data cube, which enables Online Analytical Processing (OLAP) for multi-dimensional data analysis. It facilitates faster retrieval and enhances spatial data analysis, supporting advanced data exploration and decision-making processes.
- **Open Data Standards:** This parameter emphasizes the development and enforcement of documented, reusable agreements that facilitate the publishing, access, sharing, and utilization of high-quality data within the Geospatial ecosystem. It promotes the adoption of common models, syntax, and standards for improved data interoperability.
- **Earth Observation Data Management:** This parameter focuses on developing downstream capacity for Earth Observation data, including storage, processing, and subsequent analysis at scale. It aims to enhance the management of large volumes of Earth Observation data to extract valuable insights for various applications.
- **Sensor Data Management:** This parameter involves developing the capacity to handle real-time streaming data from sensor networks, including storage, processing, and on-the-fly analytics. It enables efficient management and utilization of sensor data for real-time monitoring, analysis, and decision-making.
- **Crowdsourcing:** This parameter establishes a framework and platform for crowdsourcing data and managing the collected data effectively. It involves creating data storage, structuring, and contextualization mechanisms to leverage crowd-sourced data for various applications.

Technology



Technology is an essential and critical component within the desired Geospatial ecosystem. It plays a pivotal role in facilitating the data management components with required characteristics and implementing policy and standards frameworks, ensuring consistency and quality across the ecosystem. Technology advancements, such as automation, integrated platforms, cloud computing, connectivity solutions, and

emerging technologies like artificial intelligence and machine learning, Internet of Things (IoT), Big Data Analytics, Augmented Reality (AR), and Virtual Reality (VR) have revolutionized the Geospatial domain, enhanced its capabilities and expanded its potential applications. Within the Technology component, there are 11 key requirement parameters:

- **Current Technical Infrastructure:** This parameter will focus on assessing the existing technical infrastructure and systems to determine the requirements for seamless integration and interoperability within the Geospatial ecosystem.
- **Geospatial Automation:** This parameter focuses on the automation of Geospatial tasks such as spatial referencing, geocoding, symbolization, map production, and spatial analysis to streamline processes and improve efficiency.
- **Distributed Computing:** Establishing an interconnected network that collaboratively stores, processes, and analyzes Geospatial data across multiple nodes or servers.
- **Enterprise Cloud-based Solution:** Distributing the Geospatial data, processing, and



analysis across multiple cloud resources and services for scalability and flexibility.

- **Technology Integration:** Integrating the Geospatial data with other data sources, such as sensor networks, IoT devices, and social media, to create rich-content applications that provide users with relevant information based on location or user interests.
- **Ubiquitous Connectivity (5G/6G):** To provide a communication infrastructure capable of facilitating communication and interaction among members of the Geospatial ecosystem, enabling the transfer of substantial spatial data and sharing of computing resources.
- **Cybersecurity:** Implement mechanisms to secure access to Geospatial data, protect data in storage and during transfer, and ensure the backup and recovery of data in case of compromise. It also includes securing access to platforms and computing resources within the ecosystem.
- **Robotic Process Automation:** Incorporate Robotic Process Automation workflows to automate high volumes of repeatable tasks, improving efficiency and accuracy within the Geospatial ecosystem.
- **Quantum Computing:** Explore and incorporate Quantum Computing as a transformative technology to enhance Geospatial functionalities, accelerating optimization and machine learning processes.
- **Earth Observation:** Develop and deploy infrastructure and platforms for gathering information about the Earth's physical, chemical, and biological systems using remote sensing technologies, such as satellite-based sensors.
- **Satellite-based Positioning:** Establish infrastructure for a positioning and navigation system comprising a constellation of satellites broadcasting navigational signals, along with a network of ground and control stations for monitoring and control purposes.



Saudi Arabian National Geospatial Ecosystem Framework

The National Geospatial Center (NGC) in GEOSA will host the Saudi Arabian National Geospatial Ecosystem. In the new set up, NGC's renewed Vision, Mission, Values and Functions will serve to deliver the objectives of the Geospatial Ecosystem, as described in the previous sections.

Vision



The Saudi Arabian National Geospatial Ecosystem is the Saudi Arabian sustainable, active, and nationally relevant Geospatial Ecosystem that contributes to all activities and applications that require any element of Geospatial knowledge through an interconnected Geospatial Ecosystem of people, policies, practices, and procedures addressing the needs of the modern world.

Mission



Saudi Arabian National Geospatial Ecosystem provides a ubiquitous network of interconnected Geospatial information, products, technology, and services, enabling the delivery of knowledge-based insights to facilitate a bottom-up approach to problem-solving and decision-making.

Objectives



The goals of the Saudi Arabian National Geospatial Ecosystem go far beyond merely exchanging data. The goal is to create a network that provides powerful components, data, and services that directly address real-world problems and future what-if scenarios. At the same time, the Saudi Arabian National Geospatial Ecosystem will be able to react dynamically to changes in the sense of a system.

It is essential that parts within the Saudi Arabian National Geospatial Ecosystem can respond flexibly to changing conditions without burdening the overall system. The Saudi Arabian National Geospatial Ecosystem will work towards the following objectives:

- Create a secure, shared network based on commonly agreed-upon Geospatial principles, policies, and standards.
- Connect diverse stakeholders for Geospatial data and solutions partnerships.
- Integrate existing datasets within the Saudi Arabian Geospatial network by adopting a shared semantics-based architecture.
- Deliver knowledge-oriented analytics and solutions for real-world problems leveraging many Geospatial infrastructures and services.
- Provide advanced visualization options to experience the Ecosystem's Geospatial data and associated non-spatial attributes.
- Adopt an intelligent automated approach limiting human interventions.
- Embrace a user-centric approach to deliver personalized Geospatial products and services.



Values



Having a set of values is instrumental in shaping organizational culture, guiding behavior, and creating a foundation for long-term success. The following identified Values of the Saudi Arabian National Geospatial Ecosystem will contribute to its overall effectiveness by fostering collaboration, influencing decision-making, and building consensus both internally and externally. The foundational values of the Saudi Arabian National Geospatial Ecosystem are:

Systemic	Saudi Arabian National Geospatial Ecosystem adopts a systemic approach to the technology Ecosystem, wherein all elements - technology, actors, policies, or procedures- will be deeply interconnected and profoundly impact each other. Any shift in one aspect will result in a change in the entire system.
Dynamic	Saudi Arabian National Geospatial Ecosystem will be dynamic and continuously evolving. It will be a self-learning system based on Artificial Intelligence and Machine Learning. It will also be sensitive to external policy or process changes and adapt accordingly.
User-Centric	At the heart of Saudi Arabian National Geospatial Ecosystem is the user - government, enterprise, academic, or an ordinary person. The Ecosystem allows users better access to Geospatial-based knowledge for decision-making and problem-solving.
Knowledge-Based	Unlike present-day Geoportals or dashboards that assimilate information from various sources and deliver analytics based on it, Saudi Arabian National Geospatial Ecosystem will generate real-time insights from a wide range of such Geoportals to form a collaborative space for Geospatial knowledge.
Automated	Saudi Arabian National Geospatial Ecosystem will be an end-to-end automated system based on a self-organizing network. All data processing, assimilation, and archiving will be automated.
Secure	Safety will be a core value of the Saudi Arabian National Geospatial Ecosystem, securing data, platforms, and networks through various means.
Collaboration & Competition	Diverse actors within the Saudi Arabian National Geospatial Ecosystem will collaborate to leverage the Ecosystem and contribute to the diversity of data and processes. Here, a variety of stakeholders will be essential to maximize value. A homogenous group of actors will lose a competitive advantage.
Futuristic	Saudi Arabian National Geospatial Ecosystem will deliver will be futuristic in approach, open to changes in technology, systems, and processes, and providing solutions that can be useful yet sustainable.

Table 2: Saudi Arabian National Geospatial Ecosystem Values

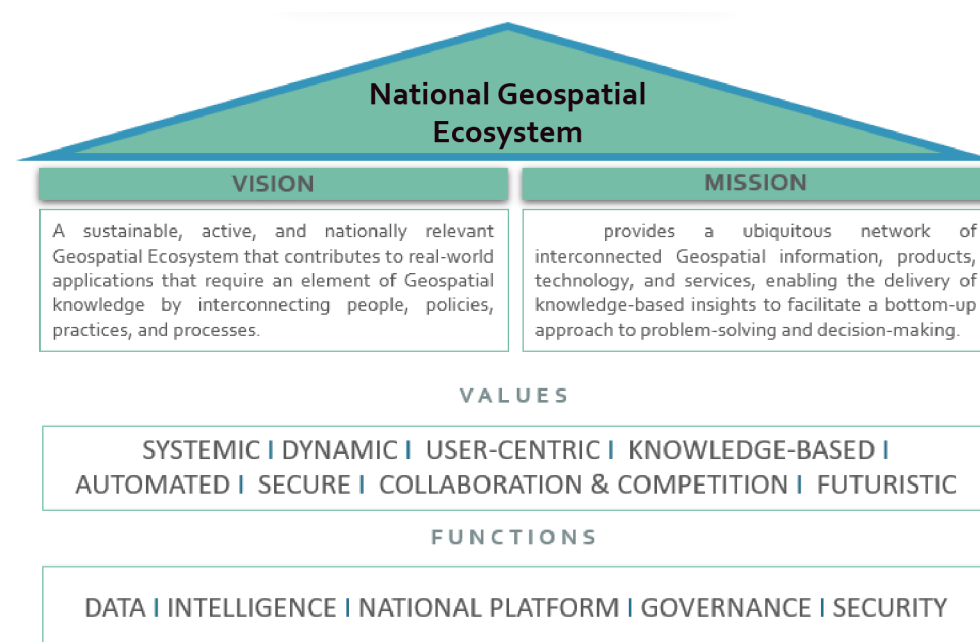


Figure 14: National Geospatial Ecosystem Vision, Mission, Values and Functions for Geospatial Ecosystem

Saudi Arabian National Geospatial Ecosystem takes a systemic approach. Though all elements, including technology, actors, policies, or procedures, will be deeply interconnected and profoundly impact each other, they will still have sufficient autonomy to adapt dynamically to changing conditions or requirements without burdening the entire system.

The Geospatial Ecosystem will use the latest full-spectrum interoperability technology and Artificial Intelligence technology. This includes applying successful software development and IT operations methods such as continuous deployment, delivery, testing, integration, and mapping to the Ecosystem. The technology will maintain a knowledge graph of related elements so that all affected artifacts within the Geospatial Ecosystem can be evaluated for necessary modifications based on any changes in the system.

The Saudi Arabian National Geospatial Ecosystem puts the user at the center of development. This user-centric approach continues to have a low barrier to entry for all participants and prioritizes the benefits of the Geospatial Ecosystem. Centralized knowledge-generation efforts will complement this user-centric approach. This work uses a central perspective on all Geospatial Ecosystem data and services, creating a knowledge base beyond bilateral insights. It will be supported by Artificial Intelligence and Machine Learning to detect patterns and insights.

In addition to the user-centric approach, the Geospatial Ecosystem will provide orthogonal services, such as cyber security, governance, legal security, and trust-building services. These services will enable a new level of collaboration and competition, as all parties can rely on the offerings technically, content-wise, and legally.

The Saudi Arabian National Geospatial Ecosystem may seem futuristic in many ways, and not all details may be apparent now. However, the Geospatial Ecosystem can continuously adopt changes to advance as a technology framework.

Functions

To effectively achieve the objectives of the Saudi Arabian National Geospatial Ecosystem, The National Geospatial Centre (NGC) will adopt new functions that will enable it to embrace the Geospatial Ecosystem concept and its associated benefits fully.

The new NGC functions have been proposed for the Geospatial Ecosystem to fulfill its responsibilities. These functions are closely linked to fulfill Saudi Arabian National Geospatial Ecosystem's mission and contribute to the Saudi Arabian Geospatial Ecosystem's growth and development.

An overview of the 5 functions is stated below:

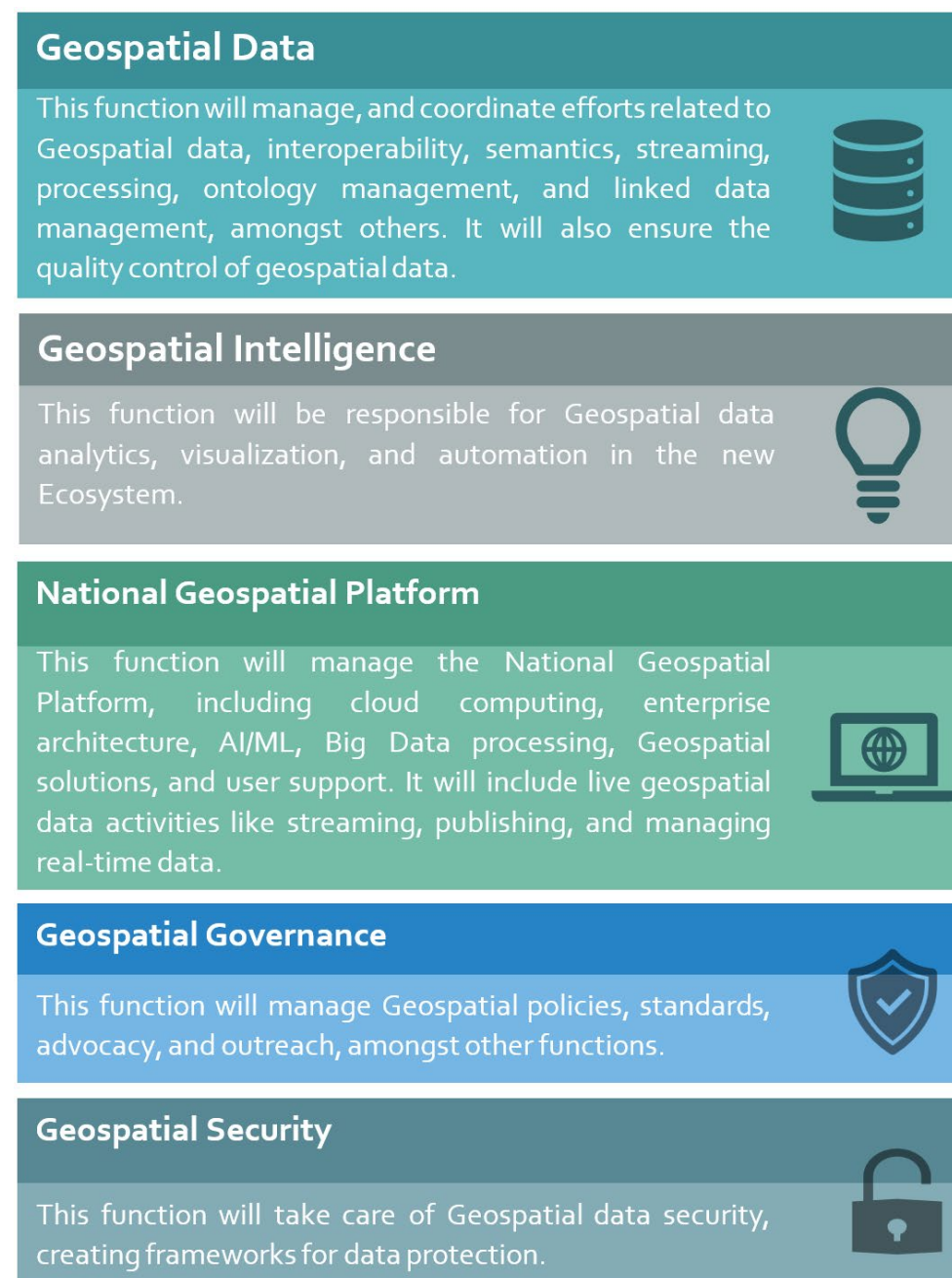


Figure 15: Functions of the Geospatial Ecosystem



Saudi Arabian National Geospatial Ecosystem Initiatives

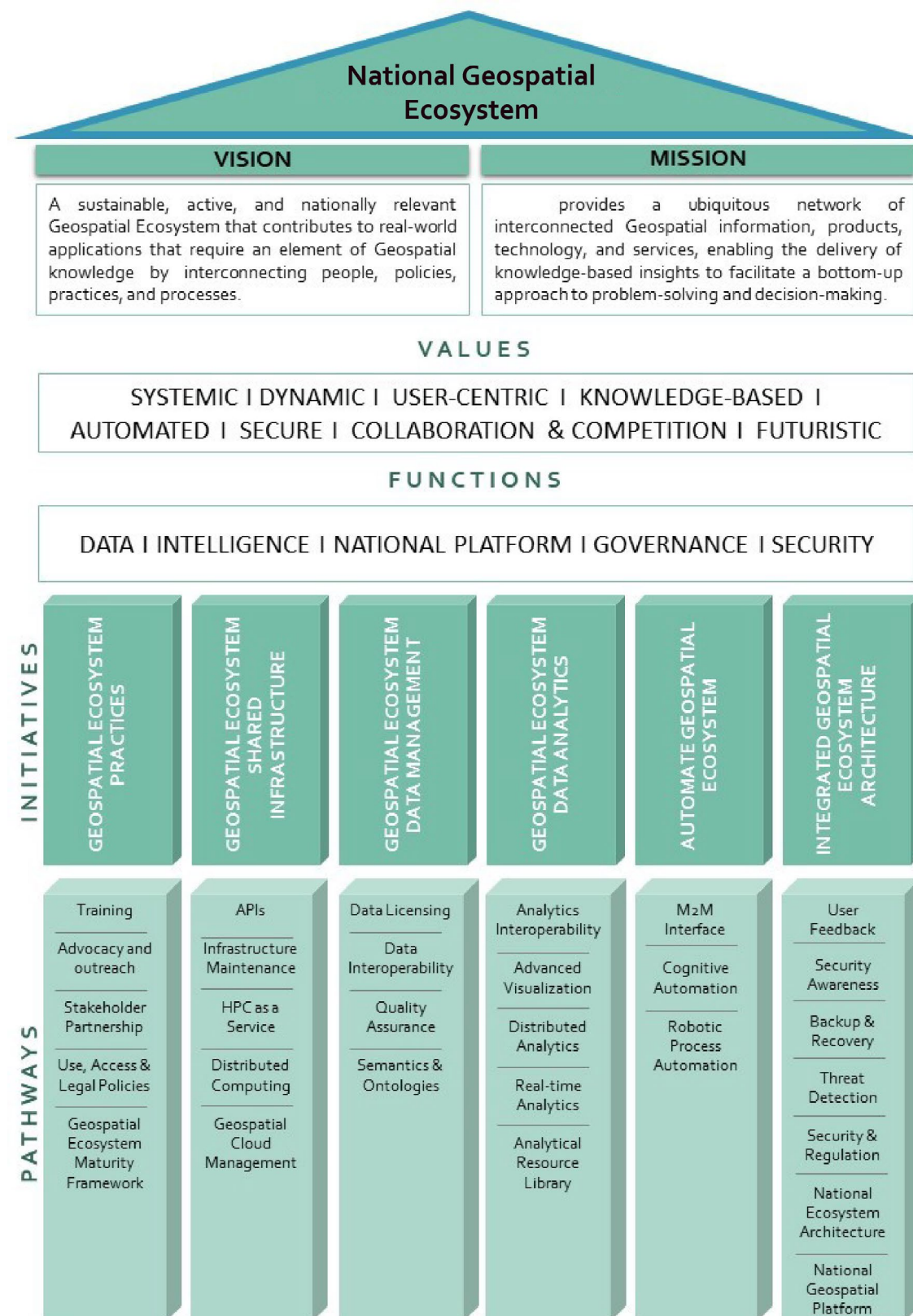


Figure 16: National Geospatial Ecosystem Vision, Mission, Values, Functions, Initiatives and Pathways for Geospatial Ecosystem



Saudi Arabian National Geospatial Ecosystem is structured around six high-level **Initiatives** with 30 associated **Pathways**, that ultimately culminate in achieving 51 **Outputs**.

The Initiatives provide a direction for the development and implementation of the Ecosystem, and at their core, address the most immediate and practical concerns related to making the data and systems currently adopted in NGC Systemic, Dynamic, User-Centric, Knowledge-Based, Automated, Secure, Collaborative, Competitive, and Futuristic.

The Saudi Arabian National Geospatial Ecosystem is based on six high-level Initiatives (Figure 19). These Initiatives collectively address all the key aspects of the Geospatial Ecosystem, ensuring that it is well-established, integrated, and functioning. Each initiative is characterized by specific objectives and expected outcomes.

Successful implementation of the six high-level **Initiatives** relies on the execution of one or more Pathways, specifically designed to attain the outcomes prescribed for each Initiative.

Pathways essentially represent a task or process to be undertaken when implementing the geospatial ecosystem around the six high-level Initiatives. Pathways would result in producing capabilities or functionalities otherwise referred to as **Outputs**. Outputs that would go on to characterize and operationalize the ecosystem.

The successful establishment and delivery of the six high-level initiatives are crucial to transform the current NGC from a data infrastructure to a national ecosystem and achieve its Mission Statement: Saudi Arabian National Geospatial Ecosystem provides a ubiquitous network of interconnected Geospatial information, products, technology, and services, enabling the delivery of knowledge-based insights to facilitate a bottom-up approach to problem-solving and decision-making.

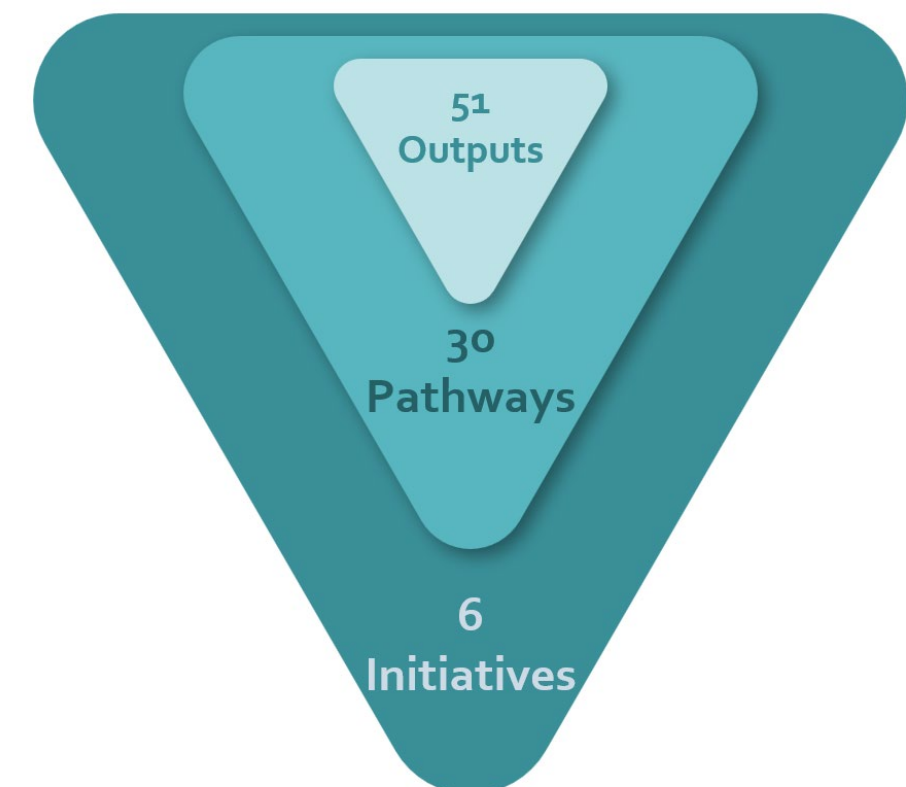


Figure 17: Breakdown of Saudi Arabian National Geospatial Ecosystem Initiatives

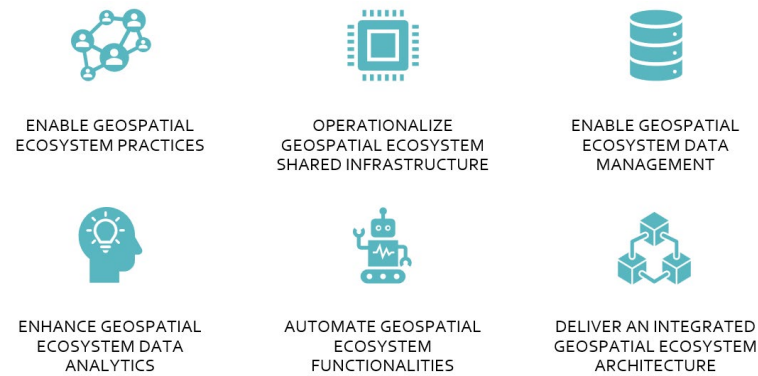


Figure 18: Saudi Arabian National Geospatial Ecosystem Initiatives

The six Initiatives, with their associated objectives and overall outcomes, are summarized in Table 3, and described in more detail below.

Initiatives	Rationale	Objectives	Outcomes
Enable Geospatial Ecosystem Practices	To adopt the best ecosystem practices and working modalities required to sustain the interconnectedness between the different stakeholders of the Geospatial ecosystem network.	To engage with the users and understand their needs to offer a highly relatable Ecosystem service while ensuring Saudi Arabian National Geospatial Ecosystem operations meet regulatory and internal process obligations.	An operational Geospatial ecosystem that productively engages various stakeholders towards an integrated approach to geo-enabled decision-making.
Operationalize Shared Geospatial Ecosystem Infrastructure	To establish and maintain a robust and scalable Saudi Arabian National Geospatial Ecosystem geospatial network of shared infrastructure.	To orchestrate Saudi Arabian National Geospatial Ecosystem technology infrastructure to support shared resource access and deliver highly available end-user applications and services.	A highly available end-user platform with an underlying technology infrastructure for Saudi Arabian National Geospatial Ecosystem.
Enable Geospatial Ecosystem Data Management	To develop comprehensive, structured, efficient Saudi Arabian National Geospatial Ecosystem data management practices with applicable policies, standards, and guidelines.	To govern and manage all aspects of Saudi Arabian National Geospatial Ecosystem data.	An integrated and contextualized Saudi Arabian National Geospatial Ecosystem data environment is developed and maintained.
Enhance Geospatial Ecosystem Data Analytics	To leverage the full potential of geospatial data and analytics within Saudi Arabian National Geospatial Ecosystem through data contextualization and ensuring analytics interoperability.	To generate valuable insights for various real-world applications and decision-making from geospatial data.	Interoperable analytical resources that enable spatially driven decision-making are developed.
Automate Geospatial Ecosystem Functionalities	To automate standalone functionalities or processes to develop capacity across both robotic process automation and cognitive automation.	To adopt ecosystem-wide automation enabled by increasingly intelligent technology and associated functionalities.	Automated processes and solutions are developed throughout Saudi Arabian National Geospatial Ecosystem.
Deliver an Integrated Geospatial Ecosystem Architecture	To ratify Saudi Arabian National Geospatial Ecosystem architecture that fosters interoperability, tying together its business, data, application, technology, and security components within the ecosystem.	To design, operationalize, and maintain the overall system architecture configured with the hardware, software, data, processes, procedures, and facilities needed to realize Saudi Arabian National Geospatial Ecosystem.	An enterprise architecture is operational and integrated throughout Saudi Arabian National Geospatial Ecosystem.

Table 3: Overview of Saudi Arabian National Geospatial Ecosystem Initiatives Rationale, Objectives and Outputs



Initiative 1: Enable Geospatial Ecosystem Practices

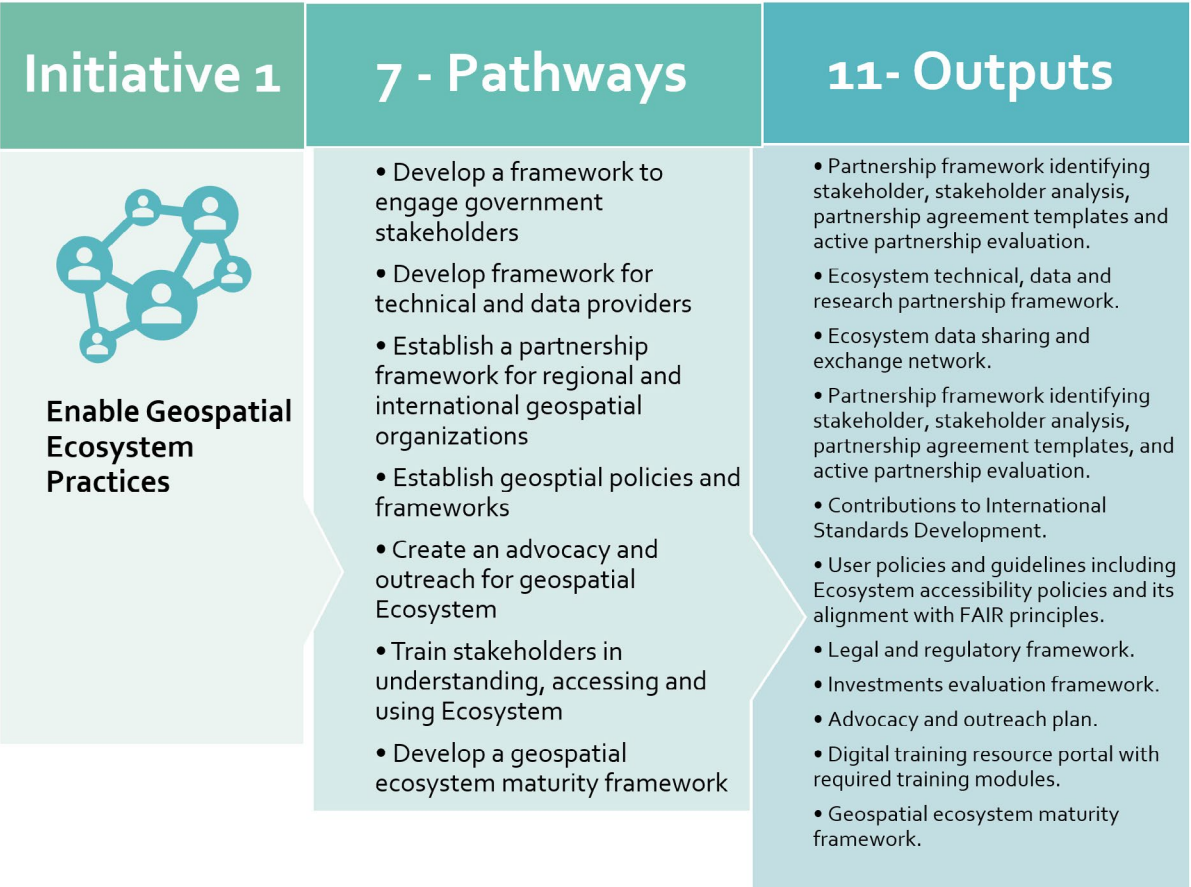


Figure 19: Initiative 1 Overview

Rationale

The Initiative “Enable Geospatial Ecosystem Practices” is driven by the imperative to embrace and uphold the best ecosystem practices and working modalities necessary to sustain the interconnectedness of stakeholders within the Geospatial ecosystem network. In today’s rapidly evolving world, geospatial information facilitates informed decision-making, resource management, disaster response, and more. It is crucial to recognize this interconnected ecosystem’s significance, optimize its operations, and ensure it operates at its full potential. This Initiative aims to create a harmonious environment where various stakeholders, from government bodies to private entities, can effectively collaborate and contribute to the growth of the Saudi Arabian National Geospatial Ecosystem. By adopting the best practices, we can ensure the seamless flow of geospatial data, information, and services, vital for holistic real-world decision making.



Objective

The initial objective of the “Enable Geospatial Ecosystem Practices” Initiative is to engage closely with Saudi Arabian National Geospatial Ecosystem users to comprehend their specific needs thoroughly. This understanding will pave the way for establishing and maintaining a highly relatable Geospatial Ecosystem that surpasses previous capabilities in addressing real-world challenges efficiently. Concurrently, the Initiative is committed to ensuring that Saudi Arabian National Geospatial Ecosystem operations remain in strict compliance with regulatory requirements and internal processes. The critical components of this objective can be summarized as follows:

1. **User-Centric Services:** Develop a deep understanding of user requirements and tailor geospatial services to meet these needs effectively.
2. **Compliance and Efficiency:** Ensure that all Saudi Arabian National Geospatial Ecosystem operations adhere to regulatory standards and are optimized for efficiency and effectiveness. Additionally, continuously assess and improve operational efficiency to maximize resource utilization and service delivery.

Initiative Outcomes

The successful execution of the “Enable Geospatial Ecosystem Practices” Initiative will result in an operational Geospatial Ecosystem that productively engages stakeholders, enabling an integrated approach to geo-enabled decision-making. This outcome envisions a geospatial landscape where:

1. **Collaboration Flourishes:** Stakeholders across sectors, regions, and organizations seamlessly collaborate, fostering a holistic approach to Geospatial information sharing and utilization.
2. **Data Accessibility and Quality:** Geospatial data is readily accessible, reliable, and conforms to international standards, ensuring its usability for diverse purposes.
3. **Efficient Decision-Making:** The Geospatial Ecosystem contributes significantly to informed decision-making across various domains, including urban planning, environmental management, disaster response, and public policy.
4. **Sustainability:** The Geospatial Ecosystem operates sustainably, with clear legal and regulatory frameworks, investment evaluation processes, and advocacy initiatives to promote its continued growth and relevance.

In summary, the “Enable Geospatial Ecosystem Practices” Initiative aims to transform the Geospatial Ecosystem into a thriving, interconnected hub where stakeholders collaborate seamlessly, data flows freely, and Geospatial services play a central role in addressing modern-day challenges. Through the Pathways outlined, it seeks to establish a foundation for a more robust and responsive geospatial landscape that serves the broader community.



Pathways

Under the “Enable Geospatial Ecosystem Practices” Initiative, the Pathways should enhance the Geospatial Ecosystem’s capabilities and practices to promote outreach, collaboration, and engagement with stakeholders at various levels, fostering a vibrant and well-coordinated geospatial ecosystem.

The following Pathways will be undertaken under this Initiative-1:

1. **Develop a partnership framework to engage government stakeholders** - This pathway aims to develop partnership frameworks to engage government stakeholders and is strategically designed to build a structured approach and engagement protocols. This initiative recognizes the critical importance of collaborating with government entities for effective data provisioning and utilization within the geospatial ecosystem. The pathway will draw upon existing partnerships within the NGC and conduct a comprehensive stakeholder analysis. This analysis will delve into the interests and involvement levels of various stakeholders, forming the basis for tailored engagement strategies.

Moreover, the pathway aims to formalize these collaborations through the establishment of partnership agreements. These agreements will be meticulously crafted to define the nature of the partnership, its scope, as well as the roles and responsibilities of each stakeholder involved. By doing so, the pathway sets the stage for fostering effective and mutually beneficial relationships with government stakeholders within the geospatial ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Partnership framework identifying stakeholder, stakeholder analysis, partnership agreement templates and active partnership evaluation.

2. **Develop framework for technical and data providers** - This pathway is dedicated to developing a partnership framework specifically tailored for non-governmental technical, data, and research providers. It will complement the previous pathway 1, which primarily focuses on engaging government stakeholders within the geospatial ecosystem. The overarching goal of this pathway-2 is to create a structured and effective engagement model, complete with protocols, for collaborating with technical, data, and research providers.

Recognizing the pivotal role these entities play in advancing the geospatial landscape, the pathway aims to leverage the latest technologies and innovations through well-defined partnerships. The pathway will thoroughly assess the technical, data, and research requirements, identify current systems and infrastructure capabilities, identify new data providers, and establish partnerships throughout the technical, data, and research network. Before issuing new Request for Interests, Request for Proposals, Service Level Agreements, or Memorandum of Understanding with relevant companies, the pathway will leverage the existing capabilities of NGC.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway is:

- i. Technical, data, and research partnership framework.
- ii. Geospatial Ecosystem data sharing and exchange network.



3. Establish a partnership framework for regional and international geospatial organizations - This pathway will focus on developing partnership framework for regional and international collaborations. It recognizes the importance of engaging with key global and regional organizations that have a direct and substantial impact on the Saudi Arabian National Geospatial Ecosystem.

The primary focus of Pathway-3 is on developing partnerships with prominent international entities, including but not limited to the United Nations Committee of Experts on Global Geospatial Information Management, the Open Geospatial Consortium, ISO/TC211, and other relevant organizations. Collaborating with these esteemed partners will provide the Saudi Arabian National Geospatial Ecosystem with an invaluable international perspective and insights. The pathway aims to create a structured framework that facilitates collaboration, information exchange, and joint initiatives with these influential organizations. By forging partnerships at the international level, the Saudi Arabian National Geospatial Ecosystem will be better equipped to address current and future real-world challenges.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Partnership framework identifying stakeholders, stakeholder analysis, partnership agreement templates, and active partnership evaluation.
- ii. Contributions to International Standards Development.

4. Establish geospatial policies and frameworks - Saudi Arabian National Geospatial Ecosystem is a new concept that requires all stakeholders' understanding and proper usage. This pathway is strategically designed to establish comprehensive use, access, and legal policies and frameworks for the Geospatial Ecosystem. This pathway focuses on creating clear frameworks and guidelines to ensure that users fully comprehend the policies and guidelines associated with Geospatial Ecosystem's use.

A core principle of Geospatial Ecosystem is to ensure it adheres to FAIR principles, i.e., Findable, Accessible, Interoperable, and Reliable, while complying with all relevant rules and regulations regarding geospatial data use. The pathway will ensure legal and regulatory compliance while analyzing the risks and return on investment in Geospatial Ecosystem. By implementing Pathway-4, the Saudi Arabian National Geospatial Ecosystem aims to create a governance structure that promotes responsible and ethical use of geospatial data. This pathway is pivotal in building trust among stakeholders and fostering a culture of compliance and accountability within the Geospatial Ecosystem framework.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. User policies and guidelines, including accessibility policies and their alignment with FAIR principles.
- ii. Legal and regulatory framework.
- iii. Investments evaluation framework.

5. Create an advocacy and outreach for geospatial Ecosystem - In the dynamic landscape of geospatial innovation, the Advocacy and Outreach Plan for the Geospatial Ecosystem stands as a strategic blueprint. The Geospatial Ecosystem requires a well-structured and targeted outreach plan to effectively communicate its purpose, advantages, and features to



all its stakeholders. Tailored for maximum impact, this pathway will be meticulously designed to convey the purpose, advantages, and distinctive features of the Saudi Arabian National Geospatial Ecosystem to diverse stakeholders by developing an advocacy and outreach plan and implementing it through a sound and effective structure.

The Advocacy and Outreach Plan is instrumental in building a positive perception of Saudi Arabian National Geospatial Ecosystem, fostering collaboration, and driving widespread adoption. It serves as a dynamic framework that adapts to the evolving needs of stakeholders and positions the Geospatial Ecosystem as a valuable asset for Saudi Arabia's geospatial community. By implementing a robust and targeted outreach strategy, this pathway ensures that Geospatial Ecosystem becomes a well-understood and embraced entity within the geospatial community and beyond. Through a structured and effective approach, the pathway aims to foster engagement, collaboration, and informed participation in the unfolding narrative of Geospatial Ecosystem's transformative journey.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Advocacy and outreach activities.

6. Train stakeholders in understanding, accessing and using Ecosystem - In the introduction of a groundbreaking concept like the Saudi Arabian National Geospatial Ecosystem, effective training and support become paramount. This pathway is dedicated to addressing the essential training needs of Geospatial Ecosystem stakeholders. It envisages a multifaceted approach, including thematic workshops that foster a deep understanding of the Geospatial Ecosystem, comprehensive manuals, and wiki pages to serve as go-to references, the development of an informative website, and the creation of tutorial videos for practical guidance. These measures collectively aim to equip every stakeholder with the requisite skills, ensuring a smooth and proficient navigation of the innovative Geospatial Ecosystem. The commitment to robust training underscores the dedication to empowering stakeholders to leverage the Geospatial Ecosystem to its fullest potential.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Digital training resource portal with required training modules.

7. Develop a geospatial ecosystem maturity framework - This pathway is dedicated to developing a Geospatial Ecosystem Maturity Framework to assess and enhance the capabilities of the Saudi Arabian National Geospatial Ecosystem. This framework will serve as a vital yardstick, annually evaluating the maturity of the Geospatial Ecosystem by considering factors like technological integration, ecosystem expansion, and collaborative partnerships. The Geospatial Ecosystem Maturity Index, a key outcome of this pathway, will not only gauge the current state of the Geospatial Ecosystem but also provide valuable indicators for improvement. This approach ensures that Saudi Arabian National Geospatial Ecosystem remains not only relevant but also continues to thrive, meeting the diverse needs of its stakeholders and fostering a culture of continuous growth within the ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- ii. Geospatial ecosystem maturity framework.



Initiative 2: Operationalize Geospatial Ecosystem Shared Infrastructure

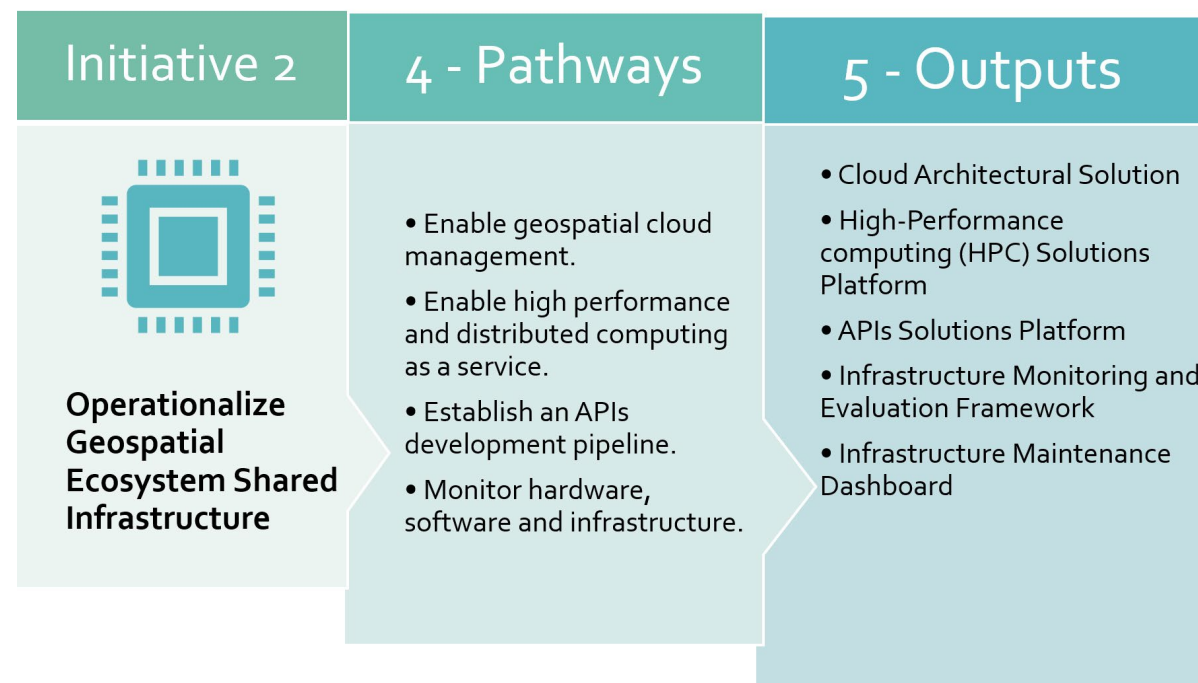


Figure 20: Initiative 2 Overview

Rationale

The Initiative ‘Operationalize Geospatial Ecosystem Shared Infrastructure’ seeks to build and operate the underlying technical infrastructure required to enable the various geospatial functionalities promoted by the proposed Saudi Arabian National Geospatial Ecosystem, ranging from spatial data access analysis to visualization and dissemination.

To enable the above-mentioned geospatial functionalities, amongst several others, there is a need for a horizontal technological solutions platform cutting across the Geospatial Ecosystem. That would enable and empower the ecosystem’s computational and networking requirements. Keeping in line with the Saudi Arabian National Geospatial Ecosystem vision of being dynamic and flexible in responding to stakeholder and user needs, the infrastructure platform would adopt a distributed approach instead of being centralized to enable shared and remote access. This platform, and by extension, the overall Initiative, would pioneer interoperability mechanisms and the development of distributed systems to support a growing number of heterogeneous geospatial end-user-ready applications and use cases.

Objective

The overall objective of this Initiative, which seeks to build and operate Saudi Arabian National Geospatial Ecosystem’s underlying technical infrastructure, would be split across three phases:

1. **Design and Develop:** The initial phase would draw from the technical needs of Geospatial Ecosystem as gathered from the Geospatial Ecosystem architecture Initiative and subsequently develop technological solutions addressing the various computational and networking needs of the ecosystem that are integrated into a single horizontal-foundational platform cutting

across Geospatial Ecosystem.

2. **Operationalize:** The secondary phase would be centered around operationalizing the Saudi Arabian National Geospatial Ecosystem technical infrastructural through implementation, which would feature procuring the required resources, establishing the necessary technological infrastructure, and integrating Geospatial Ecosystem data and application components into the established infrastructure to be able to access and leverage the infrastructure’s computing and network capabilities virtually and in real-time.

3. **Maintain and Monitor:** The tertiary phase would involve maintaining and monitoring the established technical infrastructure to ensure it is responsive to changing user and stakeholder needs regarding scalability and efficiency while exploring options for optimization and further enhancement.

Initiative Outcomes

To realize the vision of the Saudi Arabian National Geospatial Ecosystem, the Initiative aims to achieve the following key outcomes:

1. **Core Infrastructure Platform Implementation:** The foundation of Geospatial Ecosystem will involve the establishment of a robust core infrastructure platform. This platform will be designed to cater to diverse use cases and applications within the geospatial domain.
2. **Flexibility to Adapt to Ecosystem Evolution:** Acknowledging the dynamic nature of the geospatial landscape, the infrastructure will be structured to provide flexibility. This adaptability ensures that the geospatial ecosystem can seamlessly evolve to meet the emerging requirements of the ecosystem.
3. **Dynamic Resource Sharing and Benchmarking:** The implementation aims to facilitate the dynamic sharing of distributed digital resources within the ecosystem. Additionally, mechanisms for benchmarking these resources will be established, ensuring optimal utilization.
4. **Interoperability through APIs and Microservices:** Recognizing the importance of interoperability, the infrastructure will offer various levels of interoperability to clients and users. This will be achieved through the provision of well-defined APIs (Application Programming Interfaces) and microservices.
5. **User-Friendly Administration Tools:** Ensuring ease of management, the infrastructure will come equipped with user-friendly administration and management tools. These tools will empower administrators to efficiently oversee and control the various components of the geospatial ecosystem.

Pathways

To attain the objectives for the given Initiative, the following Pathways have been identified whose tasks contribute to the three phases of building and operating the Saudi Arabian National Geospatial Ecosystem technical infrastructure as described above.

The Pathways feature the provisioning of a Geospatial Ecosystem enterprise cloud solution as the core underlying platform to support the ecosystem’s functionalities regarding data, infrastructure, and applications. They are followed by the provisioning of computational resources, interoperability mechanisms to enable the integration of ecosystem components over the core cloud platform, and the essential post-operationalization process of monitoring





and maintenance.

The following are the listed Pathways under this Initiative-2:

8. **Enable geospatial cloud management** - Recognizing the pivotal role that cloud-based services play in modern computing, this pathway places a central focus on developing a robust virtual cloud environment. The primary goal is to empower the Geospatial Ecosystem with dynamic resources and functionalities hosted on the cloud infrastructure.

This includes implementing a cloud-centric approach for hosting end-user applications, ensuring scalability, and optimizing resource utilization. The aim is to facilitate seamless access to the extensive data repositories within the Geospatial Ecosystem through cloud-based services and leverage cloud capabilities to enable the dynamic sharing of infrastructure resources across the geospatial ecosystem. By embracing geospatial cloud management, the Saudi Arabian National Geospatial Ecosystem aims to enhance overall agility, accessibility, and efficiency.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Cloud Architectural Solution.

9. **Enable high performance and distributed computing as a service** - The “Enable High Performance and Distributed Computing as a Service” pathway is a critical component of the Geospatial Ecosystem. Recognizing the pivotal role of analytics in driving real-world applications and use cases, this pathway focuses on providing robust computational capacity and capability through high-performance and distributed computing services. It involves the procurement or development of High-Performance Computing (HPC) clusters, making HPC resources accessible across the entire Saudi Arabian National Geospatial Ecosystem. The HPC clusters will provide scalability to accommodate the processing demands of complex geospatial analytics.

Enabling High-Performance and Distributed Computing as a Service positions the Saudi Arabian National Geospatial Ecosystem as a cutting-edge geospatial platform capable of handling the challenges posed by extensive and intricate datasets. This initiative enhances the Geospatial Ecosystem’s analytical capabilities, contributing to its role as a dynamic and impactful geospatial ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. High-Performance computing (HPC) Solutions Platform.

10. **Establish an APIs development pipeline** - The “Establish APIs Development Pipeline” pathway aims at fostering seamless interplay and interoperability among its diverse components. Recognizing the critical role of Application Programming Interfaces (APIs) and microservices in facilitating data exchange, computational processes, and application functionalities, this pathway focuses on building a robust development pipeline. The pipeline will serve as the foundational framework for creating and deploying APIs across the entire Geospatial Ecosystem enterprise. The API development pipeline will streamline data flow across the ecosystem, promoting efficient and standardized communication. Developers will benefit from a standardized framework, accelerating the creation and deployment of APIs. Improved interconnectivity will enhance the overall interoperability of the Saudi Arabian



National Geospatial Ecosystem.

Establishing an APIs Development Pipeline is strategic for the Geospatial Ecosystem as it underpins the entire ecosystem’s ability to collaborate and operate cohesively. By providing the necessary tools for creating standardized APIs, this pathway contributes to the overall efficiency, adaptability, and collaborative potential of the Geospatial Ecosystem, ensuring it remains at the forefront of geospatial innovation.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway:

- i. APIs solutions platform.

11. **Monitor hardware, software and infrastructure** - The “Monitor Hardware, Software, and Infrastructure” pathway is a crucial pathway aimed at ensuring the continuous health, performance, and operational readiness of the shared infrastructure. As the backbone supporting various computational resources, applications, and data services, the shared infrastructure requires systematic monitoring to identify and address potential issues promptly. The main objectives of this pathway include the implementation of tools and systems for real-time monitoring of hardware, software, and overall infrastructure components. It also involves continuously evaluating the performance metrics of shared computational resources to ensure optimal functionality and conducting capacity planning exercises to anticipate future requirements and scale the infrastructure accordingly.

Monitoring the shared infrastructure directly impacts the reliability and performance of the entire Geospatial Ecosystem. Proactive identification and resolution of issues contribute to uninterrupted services, enhanced user experience, and the overall success of Saudi Arabian National Geospatial Ecosystem in delivering geospatial solutions and insights.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway:

- i. Develop infrastructure monitoring and evaluation framework, including key metrics, performance targets, and alert thresholds.
- ii. Infrastructure Maintenance Dashboard.



Initiative 3: Enable Geospatial Ecosystem Data Management

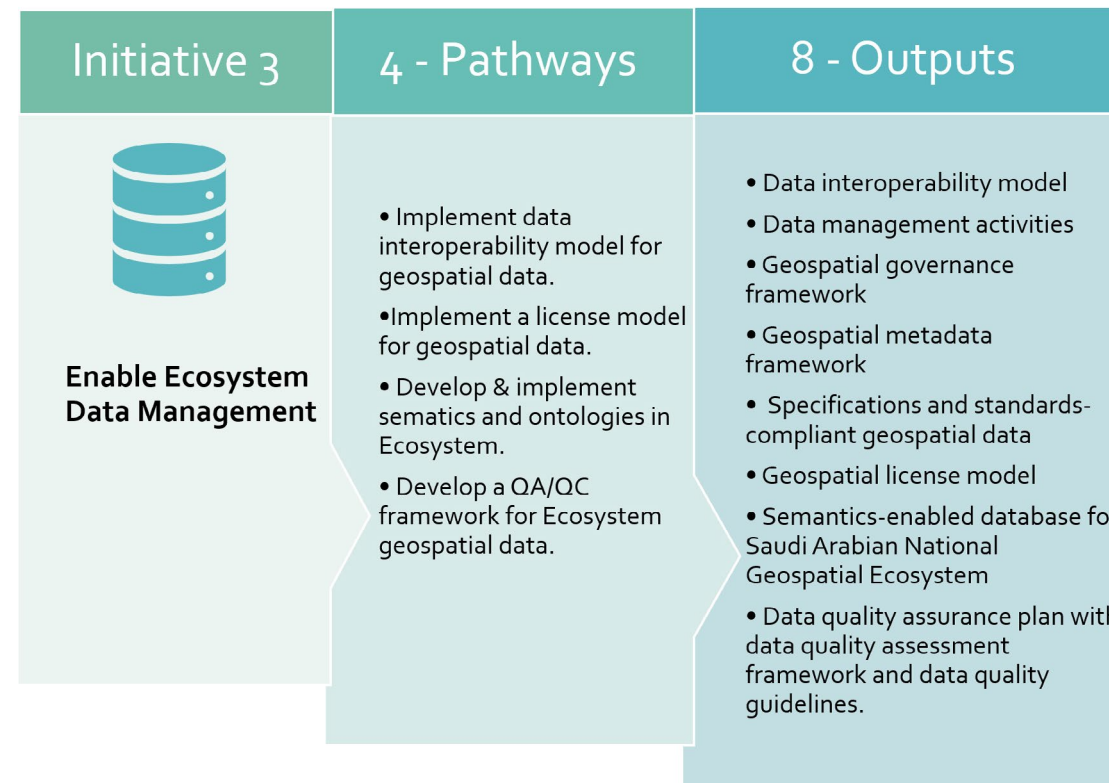


Figure 21: Initiative 3 Overview

Rationale

The “Enable Geospatial Ecosystem Data Management” Initiative is a strategic effort to establish comprehensive, efficient, and structured data management practices within Saudi Arabian National Geospatial Ecosystem. Geospatial data is a cornerstone for informed decision-making across various sectors, including government, agriculture, environment, urban planning, and disaster management. This data originates from multiple sources and arrives in diverse formats. This data must possess high quality and interoperability to be truly valuable, enabling seamless integration and analysis across the Geospatial Ecosystem systems and platforms. This Initiative strives to advance the development of data standards and guidelines that facilitate interoperability and ensure high quality.

To make accurate and timely decisions, stakeholders must access reliable, up-to-date, and well-managed geospatial data. This Initiative centers on enhancing the quality, accessibility, and sustainability of geospatial data, ensuring its value to a wide range of stakeholders. The goal is to promote the development of comprehensive, well-structured, and efficient data management practices within the Geospatial Ecosystem, incorporating relevant policies, standards, and guidelines. This includes overseeing and administering all facets of the Geospatial Ecosystem data. The objective is to create and sustain an integrated and contextualized Saudi Arabian National Geospatial Ecosystem data ecosystem.



Objective

The Initiative’s primary objective is “To govern and manage all aspects of Saudi Arabian National Geospatial Ecosystem data.” This objective is driven by the recognition that geospatial data is a critical and valuable asset that requires careful management to ensure its quality, security, accessibility, and sustainability.

The Initiative ‘Enable Saudi Arabian National Geospatial Ecosystem Data Management’ would entail objectives focused on four levels:

1. **Data Onboarding:** The initial level involves systematically gathering, verifying, and maintaining geospatial data from various sources, i.e., Data Onboarding, which ensures that relevant and accurate data is procured from external sources, such as government agencies, commercial providers, and crowd-sourced platforms.
2. **Data Enrichment:** The second level involves implementing semantic web technologies, metadata management, and ontology development. It enriches geospatial data with meaningful information and relationships, enabling better integration and enhancing the value and utilization of the data within the ecosystem.
3. **Data Interoperability:** The third level focuses on seamless data exchange and integration within the geospatial ecosystem. Data interoperability is crucial and involves developing data formats, protocols, and standards that enable different geospatial systems and applications to work together efficiently. It ensures that data can flow easily between stakeholders and across systems, enhancing collaboration and analysis.
4. **Data Quality:** The last and most crucial level is ensuring the quality, accuracy, and integrity of geospatial data is fundamental. This component is responsible for conducting thorough checks, validations, and verifications of geospatial data to identify and rectify errors, inconsistencies, and data quality issues. It seeks to ensure that the data used within Saudi Arabian National Geospatial Ecosystem is reliable and suitable for its intended purposes.

Initiative Outcomes

The successful execution of the “Enable Geospatial Ecosystem Data Management” Initiative will result in and contribute to a developed competent and structured data management practices within Saudi Arabian National Geospatial Ecosystem. The successful implementation of the Saudi Arabian National Geospatial Ecosystem data management Initiative will create a data environment where information flows seamlessly, is enriched with context, adheres to high-quality standards, and ultimately supports the diverse needs and applications of the geospatial ecosystem. This integrated and contextualized data environment is a cornerstone for the success and sustainability of Saudi Arabian National Geospatial Ecosystem.

The expected results from Saudi Arabian National Geospatial Ecosystem Data Management would ideally aim for the following:

1. **Integrated Data Environment:** Establish a systematic data onboarding process, data interoperability standards, and the implementation of semantics and ontologies.
2. **Contextualized Data:** Enriched data with meaningful context, allowing users to understand the data’s significance, relationships, and relevance.
3. **Data Quality Assurance:** Reliable and accurate integrated data environment, ensuring that decision-makers and users can trust the data for their applications and analyses.



4. **Improved Collaboration:** An integrated data environment facilitates better stakeholder collaboration.

Pathways

Under the 'Enable Geospatial Ecosystem Data Management' Initiative, the Pathways should focus on enhancing Saudi Arabian National Geospatial Ecosystem's capabilities and practices to promote efficient and structured data management practices within Saudi Arabian National Geospatial Ecosystem. They ensure that geospatial data is collected and stored, enriched with context, seamlessly integrated, and maintained at high-quality standards to support the ecosystem's various applications.

To attain the objectives for the given Initiative, the following Pathways have been identified whose Tasks contribute to the four levels towards establishing a well-structured and controlled environment for geospatial data management within Saudi Arabian National Geospatial Ecosystem as described above. The following Pathways are recommended to be undertaken under this Initiative:

12. **Implement data interoperability model for geospatial data** - Saudi Arabian National Geospatial Ecosystem is promoted as a dynamic and highly flexible platform that relies on a high level of interoperability between its various components. Data interoperability is crucial, as data serves as the primary resource that drives all other functionalities within the Geospatial Ecosystem. Therefore, this project aims to facilitate data interoperability by adopting a multi-faceted approach that addresses data governance policies to ensure quality, security, and compliance with standards; data management practices, including metadata standards, version control, and access protocols; and establish geospatial standards and specifications for consistency and compatibility; these approaches are essential prerequisites for ensuring data interoperability.

The establishment and implementation of a robust data interoperability model hold strategic significance for the Geospatial Ecosystem. It serves as the cornerstone for fostering effective collaboration, enabling advanced analytics, and facilitating informed decision-making. A seamlessly interoperable data ecosystem enhances the overall functionality of the Saudi Arabian National Geospatial Ecosystem, empowering users to extract meaningful insights from geospatial data spanning diverse domains.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Data interoperability model
- ii. Geospatial data catalogue
- iii. Geospatial governance framework
- iv. Geospatial metadata framework
- v. Specifications and standards-compliant geospatial data

13. **Implement a license model for geospatial data** - The pathway focuses on establishing clear guidelines for data licensing and usage, specifying the terms and conditions for the use of geospatial data. The pathway also includes pricing strategies that are aligned with the value of geospatial data, promoting fair and sustainable usage. Moreover, it emphasizes data sharing and protection in adherence to privacy and security standards, encouraging responsible and

efficient utilization of geospatial information.

The establishment of a robust license model for geospatial data is significant for the Saudi Arabian National Geospatial Ecosystem as it ensures responsible and ethical data usage while contributing to its economic sustainability. Well-defined licensing and pricing frameworks create an environment conducive to collaboration, innovation, and the responsible utilization of geospatial information.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway:

- i. Geospatial license model.

14. **Develop & implement semantics and ontologies** - The goal of this pathway is to improve the Geospatial Ecosystem's capabilities in managing geospatial data. To achieve this, we will focus on creating a strong framework for semantics and ontologies that will enhance the understanding and meaning of geospatial data.

The key components of this pathway are as follows: Developing a structured approach that represents the meaning of geospatial data, enabling more effective interpretation and utilization; Creating ontologies that define relationships and concepts within the geospatial domain, promoting standardized and interoperable data; Improving the ecosystem's ability to exchange and integrate geospatial data seamlessly across different systems and applications; Implementing methods for capturing and representing geospatial knowledge, enabling more intelligent and context-aware decision-making. Focusing on semantics and ontologies contributes to the development of a sophisticated and knowledge-driven geospatial ecosystem.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway:

- i. Semantics-enabled database for Geospatial ecosystem.

15. **Develop a QA/QC framework for geospatial data** - The purpose of this pathway is to enhance the overall quality, reliability, and accuracy of geospatial data within the Geospatial Ecosystem. The pathway involves creating and implementing a comprehensive Quality Assurance (QA) and Quality Control (QC) framework. This includes establishing strict procedures to validate geospatial data, ensuring that it meets predefined quality standards, implementing mechanisms for identifying errors or inconsistencies in geospatial datasets, and developing protocols for their correction. Additionally, the pathway sets up continuous monitoring mechanisms to track the quality of geospatial data over time, allowing for proactive identification and resolution of issues. By focusing on QA/QC processes, this pathway contributes to the reliability and credibility of the geospatial information provided by the Geospatial Ecosystem, building trust among users and stakeholders.

Pathway Outputs: The outputs expected from the successful implementation of this Pathway:

- i. Data quality assurance plan with data quality assessment framework and data quality guidelines.





Initiative 4: Enhance Geospatial Ecosystem Data Analytics

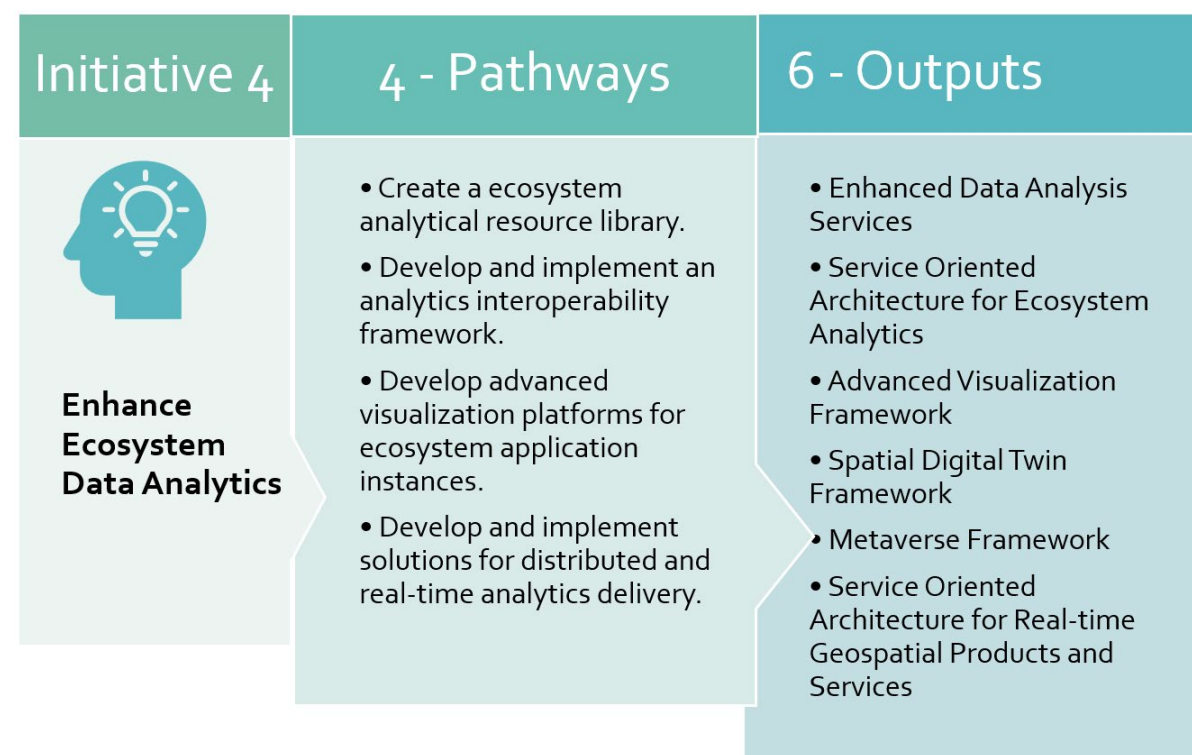


Figure 22: Initiative 4 Overview

Rationale

The primary offering of a Geospatial Ecosystem, or any instance of a Geospatial Knowledge Infrastructure is spatially driven insights to assist in real-world decision-making by users and stakeholders. Therefore, enabling a robust data analysis capability with particular emphasis on geospatial analysis a critical competency for Saudi Arabian National Geospatial Ecosystem. This will involve Artificial Intelligence, Machine Learning, Big Data Analytics, Spatial Analysis workflows dealing with raster and vector datasets, and visualization and simulation as an added medium of insights generation.

The Initiative will ensure ready-to-use data analytics, self-service solutions and capabilities, are available, accessible and interoperable throughout the ecosystem, keeping in line with the ecosystem's overarching vision of a decentralized, flexible, and dynamic platform.

Objective

The Initiative 'Enhance Geospatial Ecosystem Data Analytics' entails objectives focused on four aspects:

1. The development and deployment of analytical tools and resources, taking the form of models, software, algorithms, and workflows, capable of generating insights and performing value-addition over spatial and associated non-spatial data.
2. Consolidate analytical tools and resources in a library or catalog to enable shared and distributed access throughout the ecosystem over the latter's core cloud platform and interface.



This will involve the development of service-oriented architectures and interoperability frameworks to facilitate the use of Geospatial Ecosystem analytical resources at different stakeholder and user nodes for heterogeneous applications and use cases.

3. Provisioning real-time analytical tools and resources capable of addressing dynamic user queries and analysis requirements.
4. Developing and deploying visualization platforms and interfaces to simulate and visualize spatial concepts and scenarios. These may serve as a product or representation of the ecosystem itself, or take the form of Spatial Digital Twins, Extended Reality, or a potential Geospatial Metaverse

Initiative Outcomes

The overarching outcomes for this initiative correspond to the four critical aspects identified as a part of the objectives, and are therefore:

1. **Saudi Arabian National Geospatial Ecosystem Analytical Resources and Tools:** A consolidation of analytical resources capable of addressing the need for data-driven insights across various use cases and applications that Geospatial Ecosystem users and stakeholders fielded.
2. **Analytical Resource Interoperability:** Enabling different levels of interoperability for Geospatial Ecosystem analytical resources for their shared use and access throughout the ecosystem.
3. **Advanced Visualization Platforms and Interfaces:** Enabling the development and deployment of visualization platforms and interfaces to enhance and support Geospatial Ecosystem geospatial functionalities and use cases.
4. **Real-time Delivery of Products and Services:** Capacity to deliver analytical products and services that form a part of this Initiative in real-time, responding to dynamic user needs and interests.

Pathways

The Pathways for this Initiative, akin to the objectives, are organized across the four aspects identified above, featuring the development of Geospatial Ecosystem analytical tools and resources, followed by enabling their shared access through interoperability mechanisms, provisioning visualization as an added functionality, and the real-time and distributed delivery of all the above products and services.

The following Pathways would spearhead the 'Enhance Geospatial Ecosystem Data Analytics' Initiative:

16. **Create analytical resource library** – This pathway is essential to enable extensive data analysis capabilities, to drive value-addition to data and the generation of insights that would facilitate real-world problem-solving and decision-making. This pathway therefore focusses on building the required analytical capacities and capabilities in terms of developing data analysis tools and resources that are to be further empowered by Artificial Intelligence.

Apart from developing analytical tools and resources, the pathway would focus on integrating and consolidating analytical tools, resources, and workflows from across the ecosystem, to enable shared access and retrieval. This would feature cataloguing and benchmarking analytical



resources from across the ecosystem based on well-defined metadata, enabling targeted and easy retrieval. The pathway would feature the development of different kinds of spatial analytical tools across prescriptive, predictive, and reporting analytics. Where required the library should also host the pre-requisites of a given analytical tool such as training and testing data.

Pathway Outputs: The following outputs would be achieved through the successful implementation of this pathway:

- i) Enhanced Data Analysis Services

17. Develop & implement analytics interoperability framework - The analytical capacities and capabilities developed and consolidated as a part of the previous pathway are to be made available for use throughout the ecosystem. To be applied over heterogeneous data by its various stakeholders and users for their diverse use cases and applications. Keeping in line with the ecosystem's vision of being dynamic and flexible while enabling the generation of insights that are extensive and cut across domains in terms of value and significance. This pathway, therefore, focuses on developing solutions that would enable the interoperability and the subsequent movability of the ecosystem's analytical tools and resources. The solutions should ideally feature the development of service-oriented architectures that would facilitate interoperability of analytical resources inclusive of workflows, algorithms, and other software tools in terms of enabling their shared access and use at different nodes and platforms across the ecosystem.

Pathway Outputs: The following Outputs are expected from the successful implementation of this Pathway:

- i) Service Oriented Architecture for Ecosystem Analytics
- ii) Advanced Visualization Framework

18. Develop advanced visualization platforms for various application instances - Visualization platforms and interfaces in their different forms are essential for enabling and enhancing the perception of spatial concepts. In the case of a Geospatial Ecosystem, most of its products and services would feature the added spatial perspective. These platforms and interfaces would, therefore, serve as the instrument through which users and stakeholders' access and interact with the larger geospatial ecosystem. This pathway would, therefore, focus on setting the foundation and driving the development of visualization platforms, technologies, tools, resources, and interfaces.

Focusing beyond conventional 2D and 3D visualization techniques and onto advanced extended reality-based visualization. Apart from enhancing users' sensory augmentation, the pathway should enable the development of visualization products that offer a dynamic as opposed to static representation of real-world phenomena through means of technological paradigms such as Spatial Digital Twins and Metaverses. The pathway would feature the development of foundational frameworks and baseline tools that would facilitate the development of said visualization interfaces by stakeholders across the ecosystem to enhance the delivery of their unique use-cases and applications.

Pathway Outputs: The following Outputs are expected from the successful implementation of this Pathway:

- i) Advanced Visualization Framework



- ii) Spatial Digital Twin Framework
- iii) Metaverse Framework

19. Develop & implement solutions for real-time and distributed analytics delivery -

Apart from enabling analytical resource interoperability, the ecosystem should also benefit from the ability to perform analysis in real-time and in a distributed manner, where its execution is split across the ecosystem's computational resource network, in favor of greater optimization. This pathway will therefore focus on developing solutions towards enabling the delivery of real-time and distributed analytics within the geospatial ecosystem.

The value realization potential of geospatial products and services is directly proportional to the delivery of said products and services in real-time or near real-time. This pathway will enable the real-time delivery of ecosystem analytics by adopting a modular approach towards execution. Wherein parts of the analytical tool or software can be executed at different nodes of the ecosystem network either closer to the source of data or closer to user devices. This would ideally involve developing a service-oriented architecture that would facilitate distributed analytics execution, while also modifying individual analytical resources to be executed using a modular approach.

Pathway Outputs: The following Outputs are expected from the successful implementation of this Pathway:

- i) Service Oriented Architecture for Real-time Geospatial Products and Services.



Initiative 5: Automate Geospatial Ecosystem Functionalities

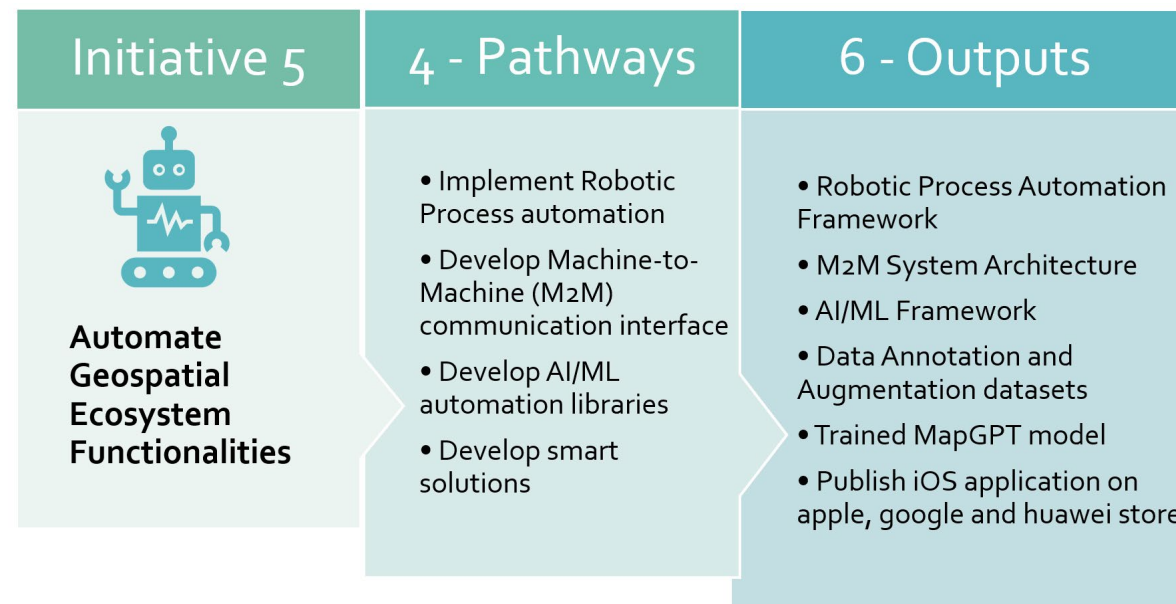


Figure 23: Initiative 5 Overview

Rationale

This initiative addresses two characteristics the Saudi Arabian National Geospatial Ecosystem aspires to emulate in its delivery of geospatial products and services, i.e., efficiency and innovation, by enabling and integrating a digital workforce in the form of automation techniques and technologies that would optimize standardized and repeatable processes and allow data to analysis at scale, considering a multitude of parameters and all the while limiting human intervention.

Envisioned as a dynamic infrastructure constantly evolving, this initiative would ensure that Saudi Arabian National Geospatial Ecosystem remains at the forefront of innovation in using Artificial Intelligence and associated paradigms for enhancing and enabling human-independent geospatial workflows

Objective

As the Initiative aspires to attain novel outcomes in terms of innovation, the objectives of this Initiative would follow a cycle of development, enablement, and implementation, wherein innovative methods and techniques are first developed and conceptualized, followed by which efforts are made toward enabling them, by procuring or developing required technology mediums and interfaces, and the subsequent phase of implementation, both proof-of-concept and wide-scale implementation.

The objectives for the given Initiative would include:

1. **Identify potential avenues for automation** of processes and functionalities throughout Saudi Arabian National Geospatial Ecosystem, in line with adopting automation as an enterprise-wide Initiative.



2. **Ascertain the value, benefits, and return on investment** to be attained as a direct result of adopting automation throughout the Saudi Arabian National Geospatial Ecosystem.
3. **Develop automation techniques and workflows** across the two automation paradigms of Robotic Process Automation and Cognitive Automation to deliver on the Geospatial Ecosystem automation needs and goals.
4. **Deploy interfaces and platforms** to develop the workflows mentioned above and enable their implementation.

Initiative Outcomes

The high-level outcomes for this initiative would correspond to the four critical aspects identified as a part of the objectives, and are therefore:

1. The establishment of automation as an enterprise-wide Initiative while standardizing the develop-enable-implement cycle to ensure its repeatability for numerous processes and functionalities that could be enhanced through automation throughout Saudi Arabian National Geospatial Ecosystem.
2. The operationalization of an infrastructural framework in terms of standards, policies, interfaces, software components, or solution-based architectures to enable the integration and adoption of a digital workforce.
3. A continuous evaluation process to monitor the benefits and return on investment attained through automation of Saudi Arabian National Geospatial Ecosystem processes and functionalities.

The consolidation of resources and tools developed throughout the Initiative to culminate in a body of knowledge for enterprise-wide automation.

Pathways

The Pathways and tasks under this Initiative are aligned with the development, enablement, and implementation of the two main automation paradigms of Robotic Process Automation and Cognitive Automation. As an Initiative that thrives on experimentation and innovation, significant emphasis has also been placed on testing and assessing the benefits of the to-be-developed functionalities as an essential task under the Pathways for this Initiative.

The following Pathways would form a part of the Initiative 'Automate Saudi Arabian National Geospatial Ecosystem Functionalities':

20. **Implement robotic process automation capabilities** - Enterprise automation would facilitate greater efficiency, effectiveness, and optimization across the geospatial ecosystem. Credentials that would be attained by enabling intelligent process automation throughout the ecosystem to limit human intervention and drive further optimization by enabling process standardization and repeatability. This pathway would, therefore, focus on enabling Robotic Process Automation (RPA) as an enterprise-wide initiative throughout the ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i) Robotic Process Automation Framework

21. **Develop a machine-to-machine communication(M2M) interface** - Limiting human



intervention as a part of automation would inadvertently mean enabling machines to interact with each other and receive and communicate tasks and outputs to each other. This pathway would, therefore, focus on enabling machine-to-machine communication throughout the ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i) M2M System Architecture

22. Develop AI/ML automation library - The previous pathways of this initiative focused on driving efficiency and process optimization through means of automation. This project should focus on developing intelligent and innovative solutions powered by Artificial Intelligence that would replicate human cognitive processing to enhance ecosystem functionalities and capabilities. The solutions so developed should be consolidated in a centralized library accessible across the ecosystem by its numerous stakeholders.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i) AI/ML framework

23. Develop smart solutions - This pathway aims to develop smart solutions, such as mobile applications and MapGPT. MapGPT is a groundbreaking artificial intelligence system designed to revolutionize the field of geospatial. MapGPT represents a cutting-edge fusion of advanced natural language processing and geospatial analysis. This innovative tool empowers users to interact with geographic data in an unprecedented way, combining the richness of language understanding with the complexity of spatial relationships. Also, the pathway will create an application to enable users to efficiently browse geospatial maps on their smartphones and tablets. The app aims to provide a seamless and high-performance experience tailored to each operating system’s specifications.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i) Annotate geospatial datasets for training, incorporating tasks such as entity recognition, spatial relationship analysis, and other relevant annotations.
- ii) Train the MapGPT model using the annotated and augmented geospatial datasets.
- iii) Develop mobile App for iO, android and huawei devices.



Initiative 6: Deliver an Integrated Geospatial Ecosystem Architecture

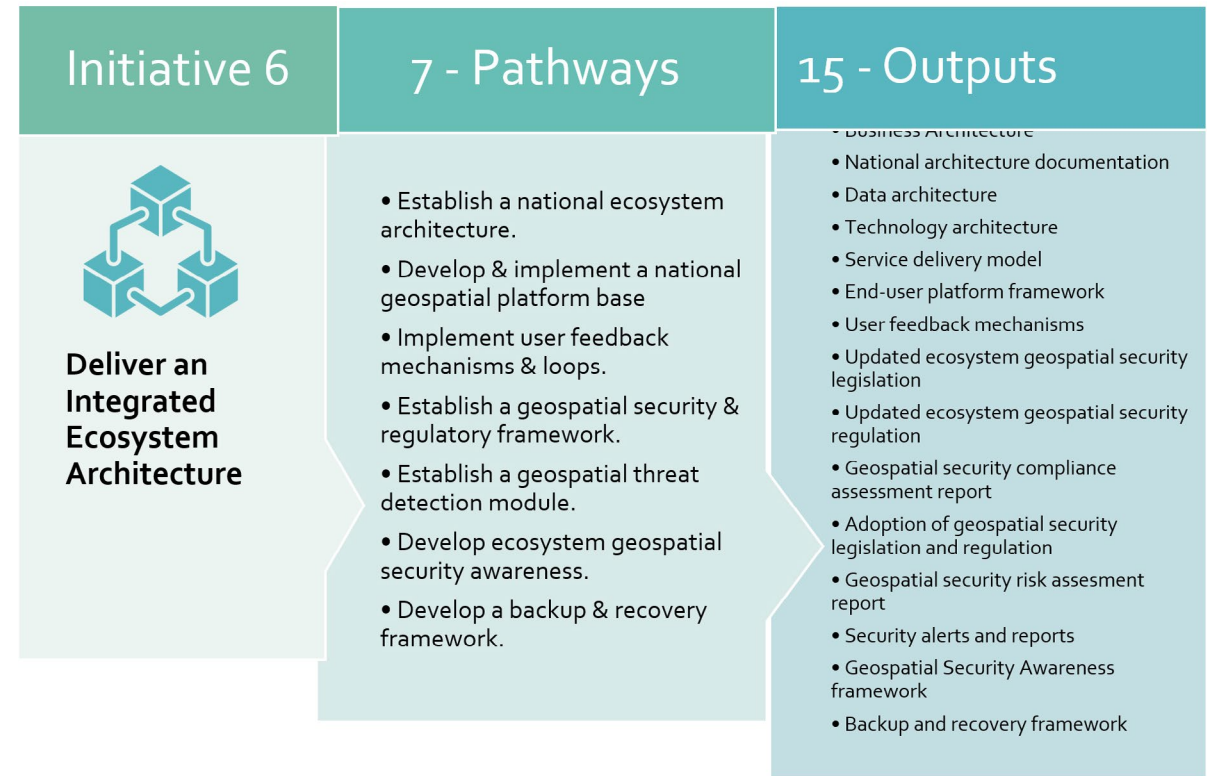


Figure 24: Initiative 6 Overview

Rationale

The “Deliver an Integrated Geospatial Ecosystem Architecture “ Initiative is multifaceted and crucial for the success and sustainability of Saudi Arabian National Geospatial Ecosystem by laying the foundation in terms of architectural framework and working modalities for enabling all the aspired functionalities and capabilities of the Geospatial Ecosystem. This Initiative aims to establish a Saudi Arabian National Geospatial Ecosystem architecture that promotes integration and interoperability by connecting its business, data, application, technology, and security components within the ecosystem.

This Initiative is critical for unlocking the complete potential of the Geospatial Ecosystem as it lays the structural foundation required to facilitate integration, interoperability, secure data exchange, and resilience, ultimately contributing to the success of Saudi Arabian National Geospatial Ecosystem’s mission and vision.

Objective

The “Deliver an Integrated Geospatial Ecosystem Architecture” Initiative aims to establish a cohesive and secure architectural framework for the Geospatial Ecosystem. This architecture should encompass the technical infrastructure and robust security, protection, and monitoring mechanisms.

The objectives of this Initiative address the six critical components of ecosystem data, business functions, technology, applications, security, and ecosystem resilience and are as follows:



1. **Establish ecosystem data architecture:** Given that all the essential functionalities and offerings of the proposed Saudi Arabian National Geospatial Ecosystem are geospatial data-driven, establishing an integrated data architecture to manage, host, and enable the decentralized movement of data across the various nodes of the ecosystem, to power applications and use cases is of the utmost importance. This would involve data modeling and the development of solution-based architecture for real-time data processing and delivery.
2. **Establish the Geospatial Ecosystem business architecture:** A business architecture would outline the Geospatial Ecosystem business outcomes while mapping them to the ecosystem's data capabilities. This would feature the initial step of identifying business outcomes and then delineating the service delivery models to be adopted to attain the outcomes.
3. **Establish the Geospatial Ecosystem technology architecture:** The Saudi Arabian National Geospatial Ecosystem technology architecture would lay the foundation for the Shared Infrastructure Initiative in identifying technical requirements throughout Saudi Arabian National Geospatial Ecosystem and conceptualizing an architectural framework capable of integrating and delivering on the said requirements. It would adopt a phased approach featuring multiple transition architectures between the current and to-be states.
4. **Enable Geospatial Ecosystem applications and user platforms:** This objective would enable the delivery of data, products, and services in the form of applications to the users and stakeholders of Saudi Arabian National Geospatial Ecosystem. It will involve the development of baseline resources and components required for developing user-oriented platforms and interfaces. A critical aspect of this objective, in line with the overall vision for Saudi Arabian National Geospatial Ecosystem, would be to keep the users and stakeholders at its center and, therefore, enable the delivery of personalized and user-centric functionalities.
5. **Ensure Geospatial Ecosystem security:** This objective would address the security component of the Geospatial Ecosystem as a high-performance and integrated network dealing with numerous and diverse stakeholders. It would involve the development of Geospatial Ecosystem security regulations and their enforcement through an operational and proactive Geospatial Ecosystem security apparatus. It would further foster a practice of secure use and access to Geospatial Ecosystem resources amongst its stakeholders.
6. **Ensure Geospatial Ecosystem resilience:** This objective seeks to ensure the resilience of the Saudi Arabian National Geospatial Ecosystem and its composite resources in any event that compromises the ecosystems' ability to function. This would entail the development of backup and recovery policies and guidelines by implementing physical mechanisms to recover the Geospatial Ecosystem and its operational capacity.

Initiative Outcomes

The initiative 'Deliver an Integrated Geospatial Ecosystem Architecture' delivers a comprehensive and holistic architectural framework for the Geospatial Ecosystem underpinned by the following overall outcomes:

1. A Geospatial Ecosystem enterprise architecture laying the foundation for operationalizing the geospatial ecosystem while addressing the critical components of business, data, and technology developed through an integrated but modular approach.
2. Enabling the delivery of user-centric and personalized products and services for users



and stakeholders through purpose-built user platforms and interfaces.

3. Ensuring the security of the Geospatial Ecosystem, its data, the underlying infrastructure, and the users and stakeholder devices and systems connected over the ecosystem by establishing a comprehensive and robust Geospatial Ecosystem security apparatus.
4. Ensuring the recovery and continuation of normal operations by the Geospatial Ecosystem in the event of a compromise by laying the framework and systems required for backup and recovery.

Pathways

The Pathways and outputs listed below comprehensively address the six components identified as a part of the Initiative's objectives. The architectural components consolidated under one master Pathway to develop the Geospatial Ecosystem national architecture, and the subsequent Pathways split across Geospatial Ecosystem user platforms and interaction, security apparatus, and backup & recovery, respectively. Following are the Pathways to be carried out as a part of this Initiative:

24. **Establish a national ecosystem architecture** - The pathway focuses on laying the foundation for enabling all the aspired functionalities and capabilities of the Saudi Arabian National Geospatial Ecosystem by developing a wholistic and integrated National Architecture for the Saudi Arabian National Geospatial Ecosystem. This national architecture promotes integration and interoperability by connecting its business, data, application, technology, and security components within the ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Business Architecture
- ii. National Architecture documentation
- iii. Data Architecture
- iv. Technology Architecture
- v. Service Delivery Model

25. **Develop & implement a national geospatial platform base** - The real-world impact and value the ecosystem seeks to generate as a part of its vision can only be realized by enabling adequate access to its functionalities and resources to its various users and stakeholders. The pathway would, therefore, focus on developing a framework comprising of baseline tools and resources for establishing user-oriented platforms, interfaces, and other services that enable easy user access to the proposed geospatial ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Baseline framework for the development of ecosystems' end-user platforms.

26. **Implement user feedback mechanisms and loops** - The proposed ecosystem offers a wide range of capabilities and functionalities. However, it is essential to note a critical difference between Spatial Data Infrastructures and the concept of a Geospatial Ecosystem. An ecosystem aims to center its platform around its users' and stakeholders' needs, challenges, and requirements. The goal therefore is to provide a more tailored experience for everyone



involved. To achieve this, this pathway would focus on gathering critical user input and accommodating them within the proposed ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this pathway:

- i. User feedback mechanisms.

27. Establish geospatial security and regulatory framework - This pathway will focus on establishing a comprehensive geospatial security and regulatory framework within the ecosystem to safeguard sensitive data, ensure compliance with relevant standards, and mitigate security risks effectively.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Geospatial security legislation & regulation
- ii. Geospatial security compliance assessment

28. Establish a geospatial threat detection module - This pathway would seek to establish a geospatial threat detection module within the ecosystem to proactively identify and respond to security threats, ensuring the integrity and security of geospatial data and services.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Geospatial security risk assessment

29. Develop Ecosystem Geospatial Security Awareness - The pathway seeks to develop and implement a Geospatial Security Awareness program as a part of ecosystem development to educate stakeholders about geospatial security risks and best practices, enhancing the overall security posture of the ecosystem.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Geospatial security awareness framework

30. Develop a backup and recovery framework - This pathway focuses on developing a comprehensive backup and recovery framework for the ecosystem, to ensure the availability and integrity of critical geospatial data and services in the event of data loss or system failures.

Pathway Outputs: The following outputs are expected from the successful implementation of this Pathway:

- i. Ecosystem backup and recovery framework

References

1. Vision 2030 Overview
vision-2030-overview.pdf (vision2030.gov.sa)
Referenced on 19th November (Page 11)
2. United Nations Committee of Experts on Global Geospatial Information Management, Report on the Eleventh session of the Committee of Experts on Global Geospatial Information Management (23 - 27 August 2021).



E_2022_46-E_C20_2021_16_E.pdf (un.org)

Referenced on 20th November 2023. (Page 15)

3. United Nations Committee of Experts on Global Geospatial Information Management, Report on the Twelfth session (3 - 5 August 2023), Meeting Document - Informal paper containing draft decisions (Part II).

20220805-12th_UN-GGIM_Informal_paper_containing_draft_decisions-Part-II.pdf.

Referenced on 20th November 2023. (Page 15)

4. Towards a sustainable geospatial ecosystem beyond SDIs.

Beyond SDI draft paper for UN GGIM (28_7_21) GS edits (1) (with SOME track changes & comments)-SC_zs-SC-changes accepted.

Referenced on 21st November 2023 (Page 18)

5. United Nations Committee of Experts on Global Geospatial Information Management, Report on the eighth session (1-3 August 2018), Economic and Social Council Official Records, 2018, Supplement No. 26.

GGIM8-report-e.pdf (un.org)

Referenced on 20 November 2023. (Page 24)

6. United Nations Committee of Experts on Global Geospatial Information Management, Report on the ninth session (7-9 August 2019), Economic and Social Council Official Records, 2019 Supplement No. 26.

GGIM9-report-e.pdf (un.org)

Referenced on 20 November 2023. (Page 24)

7. United Nations - Integrated Geospatial Information Framework part 1 Overarching Strategy document.

Microsoft Word - Part 1-IGIF Overarching Strategic Framework 24July2018.docx (un.org)

Referenced on 20 November 2023. (Page 25)

Referenced on 21st November 2023

8. Transforming our World: The 2030 Agenda for Sustainable Development.

<https://sdgs.un.org/publications/transforming-our-world-2030-agenda-sustainable-development-17981>

Referenced on 21st November 2023. (Page 34)

9. United Nations Committee of Experts on Global Geospatial Information Management, Report on the seventh session (2-4 August 2017), Economic and Social Council Official Records, 2017.

ggim-7_report_e.pdf (un.org)

Referenced on 21 November 2023. (Page 36)

10. Technical Building Blocks, Data Spaces Support Centre

Technical Building Blocks - Building Blocks - Data Spaces Support Centre (dssc.eu)

Referenced on 21 November 2023 (Page 81)

11. Common European Data Spaces, Real-time Linked Data Spaces

Common European Data Spaces - Real-time Linked Dataspaces

Referenced on 21 November 2023 (Page 84)

12. GEOSS, Group on Earth Observations

GEOSS (earthobservations.org)

Referenced on 21 November 2023 (Page 87)



13. GIGAS Methodology for comparative analysis of information
and data management systems

OGC 2010

Referenced on 21 November 2023 (Page 87)

14. OGC APIs

<https://ogcapi.ogc.org/>

Referenced on 21 November 2023 (Page 102)

